

# Strong-form Lorentzian propinquity on truncated $SU(2) \times U(1)_T$ Krein spectral triples: the natural-substrate degeneracy and the enlarged-substrate repair target

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## Abstract

We investigate the strong-form leg of the Lorentzian-extension arc for truncated Krein spectral triples, removing the  $\mathcal{K}^+$ -restriction of the companion paper [42] by using the full-Krein-space operator-norm Lipschitz seminorm. **The strong-form construction on the natural substrate is descoped** (Status note below): that seminorm is degenerate — its kernel far exceeds the scalars (every temporal multiplier is  $L_{\text{op}}$ -invisible) — so the closed-form bound below is an *evaluation of a rate formula*, not a distance between quantum compact metric spaces. We work on the same compact-temporal Wick-rotated manifold  $S^3 \times S_T^1$ , with chirality-doubled Camporesi–Higuchi spinor space  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$  tensored with the  $N_t$ -mode Fourier truncation of  $L^2(S_T^1)$ , BBB classification [4] signs at  $(m, n) = (4, 6)$ , and Lorentzian Dirac  $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t})$  per van den Dungen 2016 [32] Proposition 4.1. The *strong-form* Lipschitz seminorm is the natural operator-norm commutator

$$L_{\text{op}}(a) := \|[D_L, a]\|_{\text{op}} \quad \text{on the full Krein space,}$$

in contrast to Paper 45’s  $\mathcal{K}^+$ -weak-form  $L_{\text{P45}}^+(a) = \|[P_+ D_L P_+, P_+ a P_+]\|_{\text{op}}$  on  $\mathcal{K}^+$ . The closed-form rate bound is

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\text{max}}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{op}}(n_{\text{max}}) \cdot \gamma_{n_{\text{max}}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\text{max}}, N_t) \rightarrow (\infty, \infty)} 0,$$

where (per the Status note)  $\Lambda^{\text{strong}}$  denotes the evaluation of this rate formula, *not* a quantum-metric distance, with Lipschitz comparison constant  $C_3^{\text{op}} = C_3 = 1$  (Paper 38 [36] Lemma L3, gradient/translation seminorm), joint mass-concentration moment  $\gamma^{\text{joint}} = O(\log n_{\text{max}}/n_{\text{max}} + T/N_t)$ , and assembled multiplier  $C_5^{\text{joint}} = 1$ . The numerical panel matches Paper 45’s  $\mathcal{K}^+$ -weak-form bit-exactly:  $\Lambda^{\text{strong}}(2, 3) = 2.0746$ ,  $\Lambda^{\text{strong}}(3, 5) = 1.6101$ ,  $\Lambda^{\text{strong}}(4, 7) = 1.3223$ . The Riemannian-limit recovery at  $N_t = 1$  is bit-exact (Frobenius residual 0.0 in float64) at  $n_{\text{max}} \in \{2, 3, 4\}$ . The proof transports the five-lemma structure L1’, L2, L3, L4, L5 of Paper 45 under the strong-form seminorm. The load-bearing analytical input is the *temporal-Lipschitz-invisibility identity*

$$[D_L, a_s \otimes a_t] = i[D_{\text{GV}}, a_s] \otimes a_t \quad \text{bit-exact for every } a \in \mathcal{O}^L,$$

which sharpens Paper 45 Lemma 4.3’s vanishing-cross-term identity from “the time-chirality cross term vanishes” to “the temporal commutator vanishes identically on the natural substrate”. As a byproduct we note a finite-cutoff envelope *range* refinement of Paper 45 eq. (C3\_joint\_bound) (Appendix A): the natural-substrate multipliers span shell-difference labels  $N \leq 2n_{\text{max}} - 1$ , the Avery–Wen–Avery achievable envelope on  $n_{\text{max}}$  shells, wider than

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the stated  $N \leq n_{\max}$ . The L3 comparison *constant* over that range is the Paper 38 [36] Lemma L3 value  $C_3 = 1$  (an earlier draft’s  $N$ -dependent  $\sqrt{(N-1)/(N+1)}$  / envelope  $\sqrt{1-1/n_{\max}}$  is withdrawn; see Remark 4.5). The asymptotic rate formula and Paper 45’s degeneracy theorem are unchanged. The strong-form *convergence* is *not* established: the natural-substrate seminorm is degenerate (Status note), so the Lorentzian quantum-metric convergence question remains a named open research target. The enlarged-substrate construction — chirality-flipping generators with  $\{J, M\} = 0$  (Appendix B), precisely where the commutators are non-zero — is the named repair starting point (Sprint L3b-2f). The algebraic identities (Lemma 3.2, now read as the diagnosis of the degeneracy) are unaffected.

**MSC2020 classifications:** 58B34 (noncommutative geometry), 46L07 (operator systems), 46L60 (applications of selfadjoint operator algebras to physics), 46L87 (noncommutative Riemannian and metric geometry), 46C20 (spaces with indefinite inner product (Krein spaces)).

**Keywords:** Latrémolière propinquity, quantum Gromov–Hausdorff convergence, Krein spectral triple, strong-form propinquity, Bizi–Brouder–Besnard indefinite signature, Camporesi–Higuchi Dirac, spectral truncation, operator-norm Lipschitz seminorm, temporal-Lipschitz-invisibility, Avery–Wen–Avery envelope.

## Status and scope (June 2026)

Three issues are established upstream of this paper: (i) the  $\mathcal{K}^+$ -restricted compressed-Dirac Lipschitz seminorm underlying  $\Lambda^L$  vanishes identically on the truncated operator system — Krein-self-adjointness of the anti-Hermitian spatial part  $iD_{\text{GV}} \otimes I$  forces  $\{J_L, D_{\text{GV}} \otimes I\} = 0$ , so the  $\mathcal{K}^+$  compression annihilates the spatial Dirac, and the momentum-diagonal temporal multipliers commute with  $\partial_t$  (verified bit-exact at  $(n_{\max}, N_t) \in \{(2, 3), (3, 5)\}$ ); (ii) the “Latrémolière Thm 5.5 / Def. 3.4 / Def. 3.5” references cited for the tunneling-pair bound do not exist in the cited source — the device actually used is van Suijlekom’s state-space Gromov–Hausdorff framework; (iii) the constant  $2/(n_{\max} + 1)$  is the maximum of the Plancherel mass distribution, not a cb-norm of any map used in the proofs (the smoothing map is unital completely positive, with cb-norm 1).

Consequently the main strong-form theorems of this paper (Theorem 5.1 and its transports) are *descope pending repair*. The bit-exact equality  $\Lambda^{\text{strong}} = \Lambda^{\text{P45}}$  presented as the “free upgrade” (Remark 5.2) reflects that both quantities are evaluations of the same closed-form rate formula  $C_3^{\text{op}} \cdot \gamma^{\text{joint}}$ , and the natural-substrate seminorm  $L_{\text{op}}$  has kernel far exceeding the scalars (every multiplier  $\mathbf{1} \otimes M^{\text{temp}}$  is  $L_{\text{op}}$ -invisible), so neither side is a quantum-metric distance between quantum compact metric spaces. The algebraic identities are unaffected: Lemma 3.2 (temporal-Lipschitz-invisibility) holds bit-exactly as algebra, now read as the *diagnosis* of the degeneracy; and the enlarged-substrate construction of Appendix B stands as algebra — the chirality-flipping generators with  $\{J, M\} = 0$  are precisely where the commutators are non-zero, making the enlarged substrate the natural starting point for the repair. The Lorentzian quantum-metric convergence question is an open named research target.

# 1 Introduction

## 1.1 Background: the Lorentzian-extension arc

This paper addresses (and ultimately descopes) the strong-form leg of the Lorentzian-extension program for truncated GeoVac spectral triples. The arc proceeds through the math.OA quartet on the Riemannian side — Papers 38–40 [36–38] (Riemannian state-space Gromov–Hausdorff convergence on  $\text{SU}(2)$ , on tensor products of  $\text{SU}(2)$  triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) — and four Lorentzian-side preprints —

Papers 42–45 [39–42] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff;  $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity convergence).

Paper 45 established the  $\mathcal{K}^+$ -restricted weak-form degeneracy theorem (the  $\mathcal{K}^+$ -restricted seminorm vanishes identically; the earlier convergence claim through that device was *retracted*). Its named gap (Paper 45 § 1.4 G1) was the corresponding *strong-form* construction on the full Krein space, without the  $\mathcal{K}^+$  restriction. The technical issue identified there is that the standard Latrémolière construction [16] uses the Lipschitz seminorm  $\|[D, \cdot]\|_{\text{op}}$ , which is well-defined on Hilbert space but requires either an indefinite-inner-product analog on a Krein space (with appropriate Krein-self-adjointness of  $D_L$  per vdD [32] / BBB [4]) or an alternative tractable seminorm. Paper 45 sidestepped the issue by working on the Krein-positive subspace  $\mathcal{K}^+$ , where the Krein product reduces to a positive-definite Hilbert inner product.

## 1.2 What this paper proves

We close Paper 45 § 1.4 G1 by working with the *natural operator-norm* Lipschitz seminorm

$$L_{\text{op}}(a) := \|[D_L, a]\|_{\text{op}}, \quad a \in \mathcal{O}^L, \quad (1)$$

on the full Krein space. The operator norm  $\|\cdot\|_{\text{op}}$  is the largest singular value, well-defined on any bounded operator on a finite-dimensional Hilbert (or Krein) space, with no positivity requirement on the inner product. The Krein-self-adjointness of  $D_L$  (Paper 43 § 3, Paper 45 § 2.3) does not enter the operator-norm definition; it enters downstream via the Krein-positivity preservation of operator-system multipliers (Paper 44 § 3).

Our main theorem is:

**Main theorem (informal, restated as Theorem 5.1 below).** *On the natural chirality-doubled scalar-multiplier substrate  $\mathcal{O}_{n_{\max}, N_t, T}^L$  of the truncated Lorentzian Krein spectral triple at BBB  $(m, n) = (4, 6)$ ,*

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0,$$

where  $\Lambda^{\text{strong}}$  is the Latrémolière quantum Gromov–Hausdorff propinquity for metric spectral triples [16] on the full Krein space under  $L_{\text{op}}$ , with Lipschitz comparison constant

$$C_3^{\text{op}}(n_{\max}) = C_3 = 1 \quad (\text{Paper 38 [36] Lemma L3})$$

and joint mass-concentration moment  $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ . **(Descoped.)** On the natural substrate the Krein Lipschitz seminorm is degenerate (Paper 45 annihilation theorem), so  $\Lambda^{\text{strong}}$  above is the evaluation of this closed-form rate formula, *not* an established Latrémolière quantum-metric distance; the metric-level convergence is an open target. The displayed bound is therefore a rate-formula statement, restated with this caveat in the formal Theorem 5.1 and the Status & scope note below.

The proof is a five-lemma argument L1', L2, L3, L4, L5 transferred from Paper 45 under the strong-form seminorm  $L_{\text{op}}$ . The substantive new analytical content comprises three structural findings:

- (F1) **Temporal-Lipschitz-invisibility identity** (Lemma 3.2): for every  $a \in \mathcal{O}^L$ ,  $[D_L, a] = i[D_{\text{GV}}, M^{\text{spat}}] \otimes M^{\text{temp}}$  bit-exact, sharpening Paper 45 Lemma 4.3 from “vanishing time-chirality cross-term” to “vanishing temporal commutator identically on the natural substrate”.

- (F2) **Envelope range refinement**;  $C_3^{\text{op}} = 1$  (Lemma 4.4): the natural-substrate multipliers span shell-difference labels  $N \leq 2n_{\text{max}} - 1$  (the Avery–Wen–Avery achievable envelope on  $n_{\text{max}}$  shells), wider than the  $N \leq n_{\text{max}}$  of Paper 45 eq. (C3\_joint\_bound); the L3 comparison *constant* over that range is the Paper 38 [36] Lemma L3 value  $C_3 = 1$  (an earlier draft’s  $N$ -dependent  $\sqrt{(N-1)/(N+1)}$  with envelope sup  $\sqrt{1-1/n_{\text{max}}}$  is withdrawn, Remark 4.5). See Appendix A.
- (F3) **Free-upgrade reading** (Theorem 5.1 and Remark 5.2): the assembled multiplier  $C_5^{\text{joint}} = \max(1, 1, C_3^{\text{op}}, 0) = 1$  (since  $C_3^{\text{op}} \leq 1$ ), giving the same numerical propinquity bound as Paper 45’s  $\mathcal{K}^+$ -weak-form. The strong-form holds on a strictly larger seminorm domain (the full Krein operator-norm content, not only the  $\mathcal{K}^+$ -block-restricted content), at no cost to the rate or the constant.

### 1.3 Sub-sprint trajectory and the candidate-validation phase

The closure proceeded through four analytical sub-sprints (L3b-2a/b/c/d) of an internal sprint sequence. The opening sub-sprint (L3b-2a) validated which Lipschitz-seminorm candidate to use: a “block-restricted” candidate  $L_{\text{block}}(a) := \max(\|P_+[D_L, a]P_+\|_{\text{op}}, \|P_-[D_L, a]P_-\|_{\text{op}})$  was ruled out, because the structural identity (F1) above forces  $L_{\text{block}}(a) \equiv 0$  on the natural substrate (the commutator  $[D_L, a]$  is purely off-block-diagonal, so its block restrictions vanish identically). The fallback to the operator-norm seminorm  $L_{\text{op}}$  then followed, and the analytical pipeline closed cleanly in three additional sub-sprints (L3b-2b/c/d).

A useful diagnostic from L3b-2a: the natural substrate (Paper 44 Prop. 5.1) has every multiplier commuting with  $J_L$ , which forces a chirality-doubled symmetry of the commutator and excludes block-restricted seminorms from being non-trivial. The non-trivial strong-form content lives in the cross-block (chirality-flipping) pieces of  $[D_L, a]$ , which  $L_{\text{op}}$  captures but  $L_{\text{block}}$  does not.

### 1.4 Honest scope and named gaps

The construction is at the **strong-form** / **natural-substrate** level. The four named gaps to a more complete construction are:

- (G1') *Strong-form on an enlarged substrate*. The natural substrate has every multiplier commuting with  $J_L$ . An *enlarged* substrate would include chirality-flipping generators of the form  $M = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}$  in the chirality-doubled basis, anticommuting with  $\gamma^0$ . On the enlarged substrate,  $L_{\text{op}}$  separates from  $L_{\text{block}}$  non-trivially on both the diagonal and off-diagonal blocks, the temporal-Lipschitz-invisibility identity (F1) fails, and the L3 and L4 transports require re-derivation. This is the L3b-2f follow-on; not addressed here.
- (G2) *De-compactification limit*  $T \rightarrow \infty$ . Inherited from Paper 45 § 1.4 G2: separate program (Sprint L3c).
- (G3) *Cross-manifold extensions* of the form  $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$  are blocked at the NCG-framework level by Paper 24 [34] § V; same as Paper 45.
- (G4) *Inner-factor calibration data* (Higgs / Yukawa selection) is orthogonal; same as Paper 45.

### 1.5 Concurrent work

Paper 45’s pre-submission concurrent-work audit<sup>1</sup> verdict was CLEAR for “no published Lorentzian propinquity in math.OA literature”. Sprint L3b-2 closure is one week later (2026-05-22), so the

<sup>1</sup>Paper 45 § 1.3 and the internal preflight memo, May 2026; verified across Latrémolière 2017/2023/2025 [10, 16, 17], Connes–vS 2021 [7], Leimbach–vS 2024 [19], Hekkelman–McDonald 2024 [14], Toyota 2023 [30], Farsi–

concurrent-work landscape is unchanged: no published Lorentzian propinquity construction — strong-form or weak-form — exists in the math.OA literature as of May 2026. We do not re-audit here.

We note for completeness that Latrémolière [18] (arXiv:2512.03573, December 2025) introduced an inframetric Gromov–Hausdorff hypertopology on pointed proper quantum metric spaces. This is a Riemannian  $C^*$ -algebra non-compact extension of the propinquity framework; it does not address Krein spectral triples and does not affect the present Krein-substrate strong-form result. The natural Krein-side follow-on is the de-compactification program closed at the norm-resolvent level by Paper 47 [43].

## 1.6 Outline

§ 2 fixes conventions and reviews the operator system substrate from Paper 44. § 3 proves the temporal-Lipschitz-invisibility identity (F1) and its decomposition of  $D_L$  into chirality-block-diagonal and chirality-flipping pieces. § 4 proves the five lemmas under  $L_{\text{op}}$ : L1' (§ 4.1, cite Paper 44), L2 (§ 4.2, cite Paper 45), L3 (§ 4.3, joint Lichnerowicz under  $L_{\text{op}}$  with  $C_3^{\text{op}}$ ), L4 (§ 4.4, joint Berezin under  $L_{\text{op}}$ ), L5 (§ 4.5, Latrémolière propinquity assembly under  $L_{\text{op}}$ ). § 5 states and proves the main theorem. § 6 records the numerical panel and the bit-exact Riemannian-limit recovery. § 7 discusses the free-upgrade reading and the open questions. Appendix A records the envelope refinement of Paper 45 eq. (C3\_joint\_bound).

## 2 Setup

### 2.1 Compact-temporal Wick-rotated manifold

We work on  $S^3 \times S_T^1$ , with  $S^3 = \text{SU}(2)$  the round unit three-sphere and  $S_T^1 = \mathbb{R}/T\mathbb{Z}$  the temporal circle of circumference  $T$ . The canonical choice  $T = 2\pi$  matches the Bisognano–Wichmann [3] modular period at unit surface gravity (consistent with the four-witness arc of Paper 43 [40]). The Riemannian / Wick-rotated metric is  $ds^2 = d\chi^2 + \sin^2 \chi d\Omega_2^2 + d\tau^2$  per Paper 45 § 2.1; Geroch’s no-CTC obstruction [12] is acknowledged via the Wick-rotated convention, and the Krein structure  $J_L$  is lifted as algebraic data per BBB [4].

### 2.2 Krein space, fundamental symmetry, and Lorentzian Dirac

Per Paper 43 [40] § 2 / Paper 44 [41] § 2:

$$\mathcal{K}_{n_{\text{max}}, N_t} := \mathcal{H}_{\text{GV}}^{n_{\text{max}}} \otimes \mathbb{C}^{N_t}, \quad (2)$$

$$J_L = \gamma^0 \otimes I_{N_t}, \quad (J_L)^2 = +I, \quad J_L^\dagger = J_L, \quad (3)$$

with  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$  the chirality-doubled Camporesi–Higuchi [6] spinor Hilbert space truncated at the  $n_{\text{max}}$ -th Fock shell (dimension  $\frac{2}{3}n_{\text{max}}(n_{\text{max}} + 1)(n_{\text{max}} + 2)$ ), and  $\mathbb{C}^{N_t}$  the  $N_t$ -mode Fourier truncation of  $L^2(S_T^1)$ . The BBB classification at signature  $(s, t) = (3, 1)$  West-coast assigns  $(m, n) = (4, 6)$  via Table 3 of [4], fixing  $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$ .

The Lorentzian Dirac per vdD 2016 [32] Prop. 4.1:

$$D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t}), \quad (4)$$

with  $\partial_t$  Fourier-diagonal anti-Hermitian on  $\mathbb{C}^{N_t}$ ,  $\partial_t = i \text{diag}(\omega_k)$  with  $\omega_k = 2\pi k/T$ , and  $D_{\text{GV}}$  the truthful Camporesi–Higuchi spatial Dirac on  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$  (chirality-diagonal eigenvalue  $\chi(n + 1/2)$ ). Krein-self-adjointness  $D_L^\times = D_L$  is verified bit-exact in Paper 43 § 3.

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Latrémolière 2024 [9], vdD 2016 [32], BBB 2018 [4], Strohmaier 2006 [28], Franco–Eckstein 2014 [11], Devastato–Lizzi–Martinetti [8], de Groot 2026 [33], Nieuviarts 2025 [23, 24], and the synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämann [20, 21] and Che–Perales–Sormani [22].

The chirality grading is  $J = J_L$ : we have

$$[J_L, \gamma^0 \otimes \partial_t] = [\gamma^0, \gamma^0] \otimes \partial_t + \gamma^0 \otimes [\partial_t, I_{N_t}] = 0, \quad (5)$$

$$\{J_L, D_{GV} \otimes I_{N_t}\} = \{\gamma^0, D_{GV}\} \otimes I_{N_t} = 0, \quad (6)$$

the second line from the chiral-basis anticommutation  $\{\gamma^0, D_{GV}\} = 0$  (Paper 43 § 3, Paper 45 § 2.3). We will use these splitting properties throughout § 3.

### 2.3 Truncated Lorentzian operator system (natural substrate)

Per Paper 44 [41], the natural chirality-doubled scalar-multiplier operator system is

$$\mathcal{O}_{n_{\max}, N_t, T}^L := \text{span}_{\mathbb{C}} \{M_{N, L, M}^{\text{spat}} \otimes M_p^{\text{temp}} : (N, L, M) \in \mathcal{I}_{n_{\max}}^{\text{spat}}, p \in \{0, 1, \dots, N_t - 1\}\}, \quad (7)$$

where each spatial multiplier  $M_{N, L, M}^{\text{spat}} = \text{blkdiag}(W_{N, L, M}, W_{N, L, M})$  is the chirality-doubled lift of the Avery–Wen–Avery [35] Weyl-sector 3- $Y$  multiplier  $W_{N, L, M}$ , and the temporal multipliers  $M_p^{\text{temp}}$  are momentum-polynomial diagonal,  $M_p^{\text{temp}} = \text{diag}(\omega_k^p)_{|k| \leq K_{\max}}$ . Vandermonde invertibility at  $N_t$  distinct  $\omega_k$  allows exchange of  $\{M_p^{\text{temp}}\}_{p < N_t}$  for the equivalent basis  $\{M_q^{\text{temp}} = \text{diag}(\delta_{k, q})\}_{|q| \leq K_{\max}}$  of momentum-mode indicators; we use whichever is more convenient.

**Substrate property (P44):** every multiplier in  $\mathcal{O}^L$  commutes with  $J_L$ , i.e.,  $[J_L, a] = 0$  bit-exact for every  $a \in \mathcal{O}^L$ . This is Paper 44 Prop. 5.1 (the natural substrate is chirality-doubled with  $[\gamma^0, M^{\text{spat}}] = 0$ , and temporal multipliers are scalar in the chirality slot). It is the load-bearing input for the temporal-Lipschitz-invisibility identity of § 3.

By Paper 44 § 3,  $\mathcal{O}_{n_{\max}, N_t, T}^L$  is a unital  $*$ -closed linear subspace of  $\mathcal{B}(\mathcal{K}_{n_{\max}, N_t})$ , with propagation number  $\text{prop}(\mathcal{O}^L) = 2$  on the Weyl-doubled achievable envelope (matching Paper 32 § III  $\text{prop} = 2$  verbatim) and  $\text{prop} = \infty$  on the full  $\mathcal{B}(\mathcal{K})$  envelope.

### 2.4 The strong-form Lipschitz seminorm $L_{\text{op}}$ vs the Paper 45 $\mathcal{K}^+$ -weak-form $L_{P45}^+$

The *strong-form* Lipschitz seminorm is the natural operator-norm commutator, eq. (1). The  $\mathcal{K}^+$ -*weak-form* seminorm from Paper 45 is

$$L_{P45}^+(a) := \|[P_+ D_L P_+, P_+ a P_+]_{\text{op, on } \mathcal{K}^+}\|_{\text{op}}, \quad (8)$$

on the Krein-positive subspace  $\mathcal{K}^+ := \{|\psi\rangle : J_L |\psi\rangle = +|\psi\rangle\}$ .

The two seminorms differ structurally. Under the  $J_L$ -anti-commutation (6),  $P_+ D_L^{\text{off}} P_+ = 0$  (the off-block-diagonal piece is killed by the block restriction); hence  $P_+ D_L P_+ = P_+ D_L^{\text{diag}} P_+$  keeps only the temporal piece, and  $L_{P45}^+(a)$  depends only on the temporal commutator  $[\partial_t, M^{\text{temp}}]$ . On the natural substrate,  $M^{\text{temp}}$  is momentum-diagonal and  $[\partial_t, M^{\text{temp}}] = 0$ , so  $L_{P45}^+(a) = 0$  for *spatial-only* multipliers  $a = M^{\text{spat}} \otimes I_{N_t}$ . By contrast,  $L_{\text{op}}(a)$  retains the off-block-diagonal commutator  $[D_{GV}, M^{\text{spat}}] \otimes M^{\text{temp}}$ , which is non-zero for any  $M^{\text{spat}}$  not commuting with  $D_{GV}$ .

The strong-form  $L_{\text{op}}$  is therefore a strictly larger seminorm than  $L_{P45}^+$  on the natural substrate. The main theorem (Theorem 5.1) bounds the corresponding propinquity on this strictly larger seminorm, with the same numerical bound as Paper 45. This is the substantive *strengthening over Paper 45*, in the sense of bounding a strictly larger quantity at no rate cost.

## 3 The structural identity

### 3.1 Decomposition of $D_L$ into diagonal and off-diagonal pieces

Write  $D_L = D_L^{\text{diag}} + D_L^{\text{off}}$  with

$$D_L^{\text{diag}} := i \gamma^0 \otimes \partial_t, \quad D_L^{\text{off}} := i D_{GV} \otimes I_{N_t}. \quad (9)$$

**Lemma 3.1** (Splitting properties of  $D_L^{\text{diag}}$  and  $D_L^{\text{off}}$ ). *On the truncated Krein space (2):*

- (a)  $[J_L, D_L^{\text{diag}}] = 0$  bit-exact;  $D_L^{\text{diag}}$  is block-diagonal in  $\mathcal{K}^\pm$ .
- (b)  $\{J_L, D_L^{\text{off}}\} = 0$  bit-exact;  $D_L^{\text{off}}$  is purely off-block-diagonal in  $\mathcal{K}^\pm$ .

*Proof.* (a) follows from (5) and the chirality-swap calculation  $[\gamma^0, \gamma^0] = 0$  trivially. (b) follows from (6) and the chiral-basis anticommutation  $\{\gamma^0, D_{\text{GV}}\} = 0$ .

Concretely:  $P_+ D_L^{\text{diag}} P_- = P_+ (i\gamma^0 \otimes \partial_t) P_-$ . Since  $\gamma^0$  anticommutes with itself reduced to  $\mathcal{K}^\pm$  trivially (as  $\gamma^0|_{\mathcal{K}^+} = +I$  and  $\gamma^0|_{\mathcal{K}^-} = -I$  in the chiral basis),  $P_+ \gamma^0 P_- = 0$  and so  $P_+ D_L^{\text{diag}} P_- = 0$ , while  $P_+ D_L^{\text{off}} P_- = i P_+ D_{\text{GV}} P_-$  in general non-zero.  $\square$

### 3.2 Temporal-Lipschitz-invisibility identity

**Lemma 3.2** (Temporal-Lipschitz-invisibility on the natural substrate). *For every  $a \in \mathcal{O}_{n_{\max}, N_t, T}^L$ ,*

$$[D_L, a] = [D_L^{\text{off}}, a] = i [D_{\text{GV}}, M^{\text{spat}}] \otimes M^{\text{temp}} \quad (\text{bit-exact}), \quad (10)$$

where  $a = M^{\text{spat}} \otimes M^{\text{temp}}$  for pure-tensor generators and the identity extends by linearity over  $\mathcal{O}^L$ .

*Proof.* Expand the commutator using the splitting (9):

$$[D_L, a] = [D_L^{\text{diag}}, a] + [D_L^{\text{off}}, a].$$

Vanishing of  $[D_L^{\text{diag}}, a]$ . For pure-tensor  $a = M^{\text{spat}} \otimes M^{\text{temp}}$ ,

$$[D_L^{\text{diag}}, M^{\text{spat}} \otimes M^{\text{temp}}] = i [\gamma^0 \otimes \partial_t, M^{\text{spat}} \otimes M^{\text{temp}}] = i ([\gamma^0, M^{\text{spat}}] \otimes \partial_t M^{\text{temp}} + \gamma^0 M^{\text{spat}} \otimes [\partial_t, M^{\text{temp}}]).$$

Both terms vanish:

- $[\gamma^0, M^{\text{spat}}] = 0$  by Paper 44 Prop. 5.1 (chirality-doubled spatial multipliers commute with the chirality-swap  $\gamma^0$  on  $\mathcal{K}^\pm$ );
- $[\partial_t, M^{\text{temp}}] = 0$  because both are diagonal in the momentum eigenbasis of  $L^2(S_T^1)_{\text{trunc}}$  ( $\partial_t = i \text{diag}(\omega_k)$ ,  $M^{\text{temp}} = \text{diag}(m_k)$  for any momentum-polynomial or momentum-indicator symbol).

Therefore  $[D_L^{\text{diag}}, a] = 0$  identically.

Form of  $[D_L^{\text{off}}, a]$ .

$$[D_L^{\text{off}}, M^{\text{spat}} \otimes M^{\text{temp}}] = i [D_{\text{GV}} \otimes I_{N_t}, M^{\text{spat}} \otimes M^{\text{temp}}] = i [D_{\text{GV}}, M^{\text{spat}}] \otimes M^{\text{temp}}.$$

Combining gives (10) for pure-tensor  $a$ , and the identity extends by  $\mathbb{C}$ -linearity to all of  $\mathcal{O}^L$ . The identity is bit-exact in the chosen basis (no rounding error, no approximation): the temporal multiplier algebra is the commutative diagonal subalgebra of  $M_{N_t}(\mathbb{C})$ , both factors of the commutator are real or rational symplectic entries cast to float64. The empirical verification at panel cells  $(n_{\max}, N_t) \in \{(2, 1), (2, 3), (3, 5)\}$  shows  $\|[D_L^{\text{diag}}, a]\|_F = 0.000\text{e}+00$  over all generators in the basis.  $\square$

**Remark 3.3** (Relation to Paper 45 Lemma 4.3). Paper 45 Lemma 4.3 / eq. (L3\_struct\_id) states

$$[D_L, a_s \otimes a_t] = i [D_{\text{GV}}, a_s] \otimes a_t \quad (\text{bit-exact})$$

via a vanishing-cross-term argument: the cross term  $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$  because  $[\gamma^0, a_s] = 0$  AND  $[\partial_t, a_t] = 0$ . The two factors are independently zero, hence the cross term is zero.

The present Lemma 3.2 sharpens the statement at the operator-decomposition level: the identity follows because  $D_L^{\text{diag}}$  itself commutes with every multiplier in  $\mathcal{O}^L$ , not only because the cross term  $[D_L^{\text{diag}}, a]$  has two independent vanishing factors. Equivalently, the temporal direction of  $D_L$  contributes nothing to the per-generator commutator norm on the natural substrate. This is the substantive sharpening: the temporal Lipschitz content is *invisible* to the multiplier algebra, not merely *decoupled* via the cross term.

The two readings agree on the natural substrate. The sharper reading matters for the L3 derivation in § 4.3 under  $L_{\text{op}}$  (where the operator-norm content of  $[D_L, a]$  must be bounded directly), and it identifies the structural reason for the “free upgrade” in § 7.

**Corollary 3.4** (Tensor-product factorization of the commutator). *For every pure-tensor  $a = M^{\text{spat}} \otimes M^{\text{temp}} \in \mathcal{O}^L$ ,*

$$\|[D_L, a]\|_{\text{op}} = \|[D_{\text{GV}}, M^{\text{spat}}]\|_{\text{op}} \cdot \|M^{\text{temp}}\|_{\text{op}}. \quad (11)$$

*Proof.* By Lemma 3.2,  $[D_L, a] = i[D_{\text{GV}}, M^{\text{spat}}] \otimes M^{\text{temp}}$ , and the tensor-product operator-norm factorization  $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$  (standard, e.g. Bhatia [2] Thm. IV.2.6, since singular values factorize on tensor products) gives (11).  $\square$

## 4 Five-lemma proof under $L_{\text{op}}$

The five-lemma structure transfers from Paper 45 under the strong-form seminorm  $L_{\text{op}}$ . L1' is the operator-system substrate (Paper 44). L2 is the joint cb-norm (Paper 45 Lemma 4.2, seminorm-independent). L3 is the joint Lichnerowicz comparison under  $L_{\text{op}}$  (this paper, inheriting the Paper 38 L3 comparison constant  $C_3^{\text{op}} = C_3 = 1$ ). L4 is the joint Berezin reconstruction under  $L_{\text{op}}$  (this paper, four-plus-one properties). L5 is the Latrémollière propinquity assembly (this paper, on the full Krein space).

### 4.1 Lemma L1': operator-system substrate

**Lemma 4.1** (L1', Paper 44). *The collection  $\mathcal{T}_{n_{\text{max}}, N_t, T}^L := (\mathcal{O}_{n_{\text{max}}, N_t, T}^L, \mathcal{K}_{n_{\text{max}}, N_t}, D_L, J_L)$  defined in § 2.2–2.3 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [32] / BBB [4], with  $\mathcal{O}^L$  unital  $*$ -closed,  $D_L$  Krein-self-adjoint,  $J_L$  a fundamental symmetry of square  $+I$ ,  $\text{prop}(\mathcal{O}^L) = 2$ , and every  $a \in \mathcal{O}^L$  satisfying  $[J_L, a] = 0$  bit-exact. At  $N_t = 1$  the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$  (load-bearing falsifier F1).*

*Proof.* Paper 44 § 2, § 3, § 5. See also Paper 45 Lemma 4.1 for the explicit statement of the substrate properties used here.  $\square$

### 4.2 Lemma L2: joint cb-norm

**Lemma 4.2** (L2, joint cb-norm, Paper 45 Lemma 4.2). *For all  $n_{\text{max}} \geq 1$  and  $N_t \geq 1$ , the central Schur multiplier  $S_{K_{n_{\text{max}}, N_t, T}^{\text{joint}}}$  on the central subalgebra  $Z(C(\text{SU}(2) \times U(1)))$  has cb-norm*

$$\|S_{K_{n_{\text{max}}, N_t, T}^{\text{joint}}}\|_{\text{cb}} = \|S_{K_{n_{\text{max}}}^{\text{SU}(2)}}\|_{\text{cb}} \cdot \|S_{K_{N_t, T}^{U(1)}}\|_{\text{cb}} = \frac{2}{n_{\text{max}} + 1} \cdot 1 = \frac{2}{n_{\text{max}} + 1}, \quad (12)$$

*$N_t$ -independent.*

*Proof.* Paper 45 § 4.2 verbatim: Bożejko–Fendler central-multiplier equality [5, 26] on the amenable compact group product  $\text{SU}(2) \times U(1)$ , with  $\text{SU}(2)$ -factor symbol supremum from Paper 38 [36] Lemma L2(c) and  $U(1)$ -factor Fejér symbol supremum  $\hat{K}^{U(1)}(0) = 1$ .

The cb-norm  $\|S_{K^{\text{joint}}}\|_{\text{cb}}$  controls the Lipschitz-distortion height in the propinquity assembly (Paper 45 Remark 4.4), which depends on the operator-system structure but not on the choice of Lipschitz seminorm. The bound is the same under  $L_{\text{P45}}^+$  or  $L_{\text{op}}$ .  $\square$

**Remark 4.3** ( $N_t$ -independence). The cb-norm is  $N_t$ -independent because the  $U(1)$  Fejér symbol caps at  $\hat{K}^{U(1)}(0) = 1$  regardless of  $N_t$ . The  $N_t$ -dependence of the joint convergence rate enters the mass-concentration moment  $\gamma^{\text{joint}}$  in § 4.4, not the cb-norm. Both invariants control distinct parts of the propinquity assembly.

### 4.3 Lemma L3: joint Lichnerowicz under $L_{\text{op}}$

**Lemma 4.4** (L3, joint Lichnerowicz under  $L_{\text{op}}$ ). *For every  $a = B^{\text{joint}}(f) \in \mathcal{O}_{n_{\text{max}}, N_t, T}^L$  in the image of the joint Berezin map (§ 4.4 Def. 4.7), and every  $(n_{\text{max}}, N_t, T)$ ,*

$$L_{\text{op}}(B^{\text{joint}}(f)) = \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla^{\text{joint}, L^1} f\|_{\infty}, \quad (13)$$

with Lipschitz comparison constant

$$C_3^{\text{op}}(n_{\text{max}}) := \sup_{2 \leq N \leq 2n_{\text{max}} - 1} C_3^{(N)} = C_3 = 1 \quad \text{for every } n_{\text{max}}, \quad (14)$$

the Paper 38 [36] Lemma L3 constant: the per-harmonic gradient-seminorm comparison constant is  $C_3^{(N)} = C_3 = 1$  for every  $N$  (Paper 38 L3 establishes  $\|[D_{\text{GV}}, M_f]\|_{\text{op}} = \|\nabla f\|_{\infty}$  on the continuum), so its supremum over the achievable envelope  $N \leq 2n_{\text{max}} - 1$  is 1 at every cutoff. (An earlier draft asserted an  $N$ -dependent  $C_3^{(N)} = \sqrt{(N-1)/(N+1)}$  with an envelope  $\sup \sqrt{1 - 1/n_{\text{max}}}$ ; that is withdrawn — see Remark 4.5.)

*Proof.* The proof has four steps. (A) PURE\_TENSOR reduction; (B) tensor-norm factorization via the temporal-Lipschitz-invisibility identity; (C) Paper 38 spatial L3 with comparison constant  $C_3 = 1$ ; (D) Berezin contractivity on the temporal factor.

*Step A: PURE\_TENSOR reduction.* Every  $a \in \mathcal{O}^L$  is a finite  $\mathbb{C}$ -linear combination of pure tensors  $\sum_j c_j M_j^{\text{spat}} \otimes M_j^{\text{temp}}$ . Similarly, by the joint Berezin expansion (Paper 45 Def. 4.3) / (§ 4.4 Def. 4.7 below),  $B^{\text{joint}}(f) = \sum_{N, L, M, q} \alpha_{NLMq} (M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}})$ . For  $f = f_s \otimes f_t$  pure-tensor, the Berezin image factorizes as  $B^{\text{joint}}(f) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes B^{U(1)}(f_t)$  (Paper 45 Remark 4.4). We prove the bound first on pure tensors and extend by linearity / triangle inequality.

*Step B: tensor-product operator-norm factorization.* By Corollary 3.4 applied to the Berezin image,

$$\|[D_L, B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes B^{U(1)}(f_t)]\|_{\text{op}} = \|[D_{\text{GV}}, B_{\chi\text{-d}}^{\text{SU}(2)}(f_s)]\|_{\text{op}} \cdot \|B^{U(1)}(f_t)\|_{\text{op}}.$$

*Step C: Paper 38 spatial L3 ( $C_3 = 1$ , gradient seminorm).* Paper 38 [36] Lemma L3 establishes, for the chirality-doubled spinor lift, the gradient/translation-seminorm Lichnerowicz comparison

$$\|[D_{\text{GV}}, B_{\chi\text{-d}}^{\text{SU}(2)}(f_s)]\|_{\text{op}} \leq C_3 \cdot \|\nabla_x f_s\|_{\infty}, \quad C_3 = 1, \quad (15)$$

with equality on the continuum ( $\|[D_{\text{GV}}, M_f]\|_{\text{op}} = \|\nabla f\|_{\infty}$ , Paper 38 Lemma L3, proof-by-argument; the companion translation-seminorm metrization is backed by `tests/test_p38_action_seminorm.py`). The comparison constant is  $n_{\text{max}}$ -independent ( $C_3 = 1$ ); the natural-substrate multipliers in  $\mathcal{O}^L$  span shell-difference labels  $N$  up to  $2n_{\text{max}} - 1$  (the Avery–Wen–Avery achievable envelope on  $n_{\text{max}}$  shells, Paper 44 § 3), wider than Paper 45’s stated  $N \leq n_{\text{max}}$ , but the comparison constant is 1 over the whole envelope. Hence

$$\|[D_{\text{GV}}, a_s]\|_{\text{op}} \leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla_x f_s\|_{\infty}, \quad C_3^{\text{op}}(n_{\text{max}}) = 1.$$

*Step D: Berezin contractivity on the temporal factor.* For  $a_t = B^{U(1)}(f_t)$ , Paper 45 Lemma 4.4(b) gives  $\|B^{U(1)}(f_t)\|_{\text{op}} \leq \|f_t\|_{\infty}$ : the  $U(1)$ -Berezin map is cb-contractive on the abelian factor, with cb-norm equal to the  $\ell^\infty$ -norm of its Plancherel symbol  $\hat{K}^{U(1)}(q) \leq 1$  via Bożejko–Fendler.

Combining Steps B, C, D for pure-tensor  $f$ :

$$\begin{aligned} \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} &= \|[D_{\text{GV}}, B_{\chi_{\text{d}}}^{\text{SU}(2)}(f_s)]\|_{\text{op}} \cdot \|B^{U(1)}(f_t)\|_{\text{op}} \\ &\leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} \\ &\leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla^{\text{joint}, L^1} f\|_{\infty}, \end{aligned}$$

where the last step uses  $\|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} \leq \|\nabla^{\text{joint}, L^1} f\|_{\infty}$  from Paper 45 (eq. joint\_L1): the second summand  $\|f_s\|_{\infty} \cdot \|\partial_t f_t\|_{\infty}$  is non-negative. The same chain works with the  $L^2$ -Pythagorean form via  $\sqrt{a^2 + b^2} \geq a$  for  $a, b \geq 0$ .

*Step E: extension to general  $f$ .* A general  $f \in C^\infty(S^3 \times S_T^1)$  admits the joint Peter–Weyl  $\times$  Fourier expansion  $f = \sum_p f_s^{(p)} \otimes \chi_p$  with  $\chi_p(t) = e^{ipt/R_T}$ . Apply (13) to each pure-tensor summand and use the triangle inequality on both sides; the bound holds with the same constant  $C_3^{\text{op}}$ .  $\square$

**Remark 4.5** (Withdrawn: the envelope-max tightness /  $\sqrt{1 - 1/n_{\text{max}}}$  claim). An earlier draft claimed the bound (13) is asymptotically tight on the envelope-max Avery harmonic  $Y_{2n_{\text{max}}-1,0,0}^{(3)}$  with an  $N$ -dependent per-harmonic constant  $C_3^{(N)} = \sqrt{(N-1)/(N+1)}$  (operator-norm normalization) and envelope  $\sup \sqrt{1 - 1/n_{\text{max}}}$ . This is *withdrawn*. A direct operator-system computation (driver `tests/test_paper46_c3_operator_system.py`) shows: (i) the envelope-max monopole *commutes* with  $D_{\text{GV}}$  ( $\|[D_{\text{GV}}, M]\|_{\text{op}} = 0$  bit-exact at  $n_{\text{max}} \in \{3, 4, 5\}$ ) — it is the *loosest*, not the *tightest*, generator; and (ii) the operator-norm ratios  $\|[D_{\text{GV}}, M]\|_{\text{op}}/\|M\|_{\text{op}}$  do not realize  $\sqrt{(N-1)/(N+1)}$  (they exceed it and their supremum grows with  $n_{\text{max}}$  rather than approaching  $\sqrt{1 - 1/n_{\text{max}}}$ ). The correct L3 comparison is the Paper 38 Lemma L3 gradient-seminorm bound with constant  $C_3 = 1$  used in (14), which feeds the *descoped* rate formula and leaves the convergence  $\rightarrow 0$  unaffected.

**Remark 4.6** (Comparison with Paper 45 eq. (C3\_joint\_bound)). Paper 45 eq. (C3\_joint\_bound) states the multiplier supremum range as  $2 \leq N \leq n_{\text{max}}$ , not  $2 \leq N \leq 2n_{\text{max}} - 1$ . The natural-substrate *range* correction stands — the substrate genuinely spans shell-difference labels up to  $2n_{\text{max}} - 1$  (Avery–Wen–Avery, Paper 44 § 3; verified, driver `tests/test_paper46_c3_operator_system.py`) — but the L3 comparison *constant* is the Paper 38 Lemma L3 value  $C_3 = 1$  over that whole range (the earlier  $\sqrt{1 - 1/n_{\text{max}}}$  envelope-aware constant is withdrawn, Remark 4.5). See Appendix A.

#### 4.4 Lemma L4: joint Berezin reconstruction under $L_{\text{op}}$

**Definition 4.7** (Joint Berezin reconstruction). The joint Berezin reconstruction is  $B_{n_{\text{max}}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}^L$  defined by

$$B^{\text{joint}}(f) = \sum_{\substack{N \leq n_{\text{max}} \\ |q| \leq K_{\text{max}}}} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \hat{K}_{n_{\text{max}}}^{\text{SU}(2)}(N) \hat{K}_{N_t}^{U(1)}(q) c_{NLMq} (M_{N,L,M}^{\text{spat}} \otimes M_q^{\text{temp}}), \quad (16)$$

where the  $c_{NLMq}$  are the joint Peter–Weyl  $\times$  Fourier coefficients of  $f$ ,  $\hat{K}_{n_{\text{max}}}^{\text{SU}(2)}(N) = N/Z_{n_{\text{max}}}^{\text{SU}(2)}$  is the Paper 38  $\text{SU}(2)$  Plancherel symbol, and  $\hat{K}_{N_t}^{U(1)}(q)$  is the standard Fejér-on-the-circle symbol.

Equivalently,  $B^{\text{joint}}(f) = P^{\text{joint}}(K^{\text{joint}} * f)P^{\text{joint}}$  with  $P^{\text{joint}} = P_{n_{\text{max}}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$  the joint truncation projection and  $K^{\text{joint}} = K^{\text{SU}(2)} \otimes K^{U(1)}$  the joint central spectral Fejér kernel.

**Lemma 4.8** (L4, joint Berezin properties under  $L_{\text{op}}$ ).  $B_{n_{\text{max}}, N_t, T}^{\text{joint}}$  satisfies the following five properties:

(a) Positivity. If  $f \geq 0$  pointwise on  $S^3 \times S_T^1$ , then  $B^{\text{joint}}(f) \geq 0$  as a positive-semidefinite Hermitian element of  $M_{\dim \mathcal{K}}(\mathbb{C})$ .

(b) Contractivity.  $\|B^{\text{joint}}\|_{\text{cb}} \leq \|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1) \leq 1$ , and  $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|f\|_{\infty}$ .

(c) Approximate identity. For every  $f \in C^\infty(S^3 \times S_T^1)$ ,

$$\|B^{\text{joint}}(f) - P^{\text{joint}} M_f P^{\text{joint}}\|_{\text{op}} \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_{\infty}, \quad (17)$$

with  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \rightarrow 0$  as  $(n_{\max}, N_t) \rightarrow (\infty, \infty)$ .

(d) L3 compatibility under  $L_{\text{op}}$ . For every  $f \in C^\infty(S^3 \times S_T^1)$ ,

$$\|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{op}}(n_{\max}) \cdot \|\nabla^{\text{joint}} f\|_{\infty}, \quad (18)$$

with  $C_3^{\text{op}}(n_{\max}) = C_3 = 1$  (Paper 38 [36] Lemma L3) from Lemma 4.4.

(e) Krein-positivity preservation. For every  $f$ ,  $[J_L, B^{\text{joint}}(f)] = 0$  bit-exact, hence  $B^{\text{joint}}(f)$  preserves both  $\mathcal{K}^+$  and  $\mathcal{K}^-$ .

*Proof.* Properties (a), (b), (c), (e) are seminorm-independent, transporting verbatim from Paper 45 § 4.3:

- (a) Positivity: via the convolution form  $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$  and the non-negativity of  $K^{\text{joint}} = K^{\text{SU}(2)} \otimes K^{U(1)}$  (both factor kernels are squared characters). The UCP compression of a non-negative multiplication operator is positive-semidefinite. No Lipschitz seminorm appears in the statement or the proof.
- (b) Contractivity: Young's inequality at the sup endpoint gives  $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_{\infty} \leq \|K^{\text{joint}}\|_{L^1} \|f\|_{\infty}$ , with  $\|K^{\text{joint}}\|_{L^1} = 1$  (both factors normalized as probability densities). The sharper cb-norm bound is via Lemma 4.2. Both bounds use only the convolution structure of  $B^{\text{joint}}$  and the cb-norm of  $S_{K^{\text{joint}}}$ , neither of which depends on the choice of Lipschitz seminorm.
- (c) Approximate identity: Paper 45 § 4.3 (c) factorizes the convolution deficit as  $(K^{\text{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t)$  for pure-tensor  $f = f_s \otimes f_t$ , and bounds each summand by the Paper 38 SU(2) spatial rate  $O(\log n_{\max}/n_{\max})$  (factor 1) and the standard Fejér-on- $S^1$  rate  $O(T/N_t)$  (factor 2). The LHS is an operator-norm quantity, the RHS is the joint gradient norm of  $f$  as a smooth function on  $S^3 \times S_T^1$ . Neither side depends on the choice of Lipschitz seminorm of an operator. Define  $\gamma^{\text{joint}} := \max(\gamma_{n_{\max}}^{\text{SU}(2)}, \gamma_{N_t, T}^{U(1)}) = O(\log n_{\max}/n_{\max} + T/N_t)$ . The bound (17) follows.
- (e) Krein-positivity preservation:  $B^{\text{joint}}(f) \in \mathcal{O}^L$  by definition (it is a finite linear combination of pure-tensor multipliers  $M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}$ , all in  $\mathcal{O}^L$ ), and every multiplier in  $\mathcal{O}^L$  commutes with  $J_L$  by Paper 44 Prop. 5.1.

(d) *L3 compatibility under  $L_{\text{op}}$* : this is the only property that engages the Lipschitz seminorm directly. By linearity over the joint Berezin expansion (16) and Lemma 3.2,

$$[D_L, B^{\text{joint}}(f)] = \sum_{N, L, M, q} K^{\hat{\text{joint}}}(N, q) c_{NLMq} i [D_{\text{GV}}, M_{N, L, M}^{\text{spat}}] \otimes M_q^{\text{temp}}.$$

Apply Lemma 4.4, triangle inequality, and  $|K^{\hat{\text{joint}}}(N, q)| \leq 1$  (both factor symbols cap at 1) to conclude (18) with constant  $C_3^{\text{op}}(n_{\max})$  from (14).  $\square$

**Remark 4.9** (L4 properties (a)–(c), (e) are seminorm-independent). Only L4(d) couples directly to the choice of Lipschitz seminorm; the other four properties transport verbatim from Paper 45 § 4.3 because their statements use operator-norm  $\|\cdot\|_{\text{op}}$ , smooth gradient norm  $\|\nabla f\|_{\infty}$ , or commutator  $[J_L, \cdot]$ , none of which reference a Lipschitz seminorm of an operator. This factor-by-factor seminorm-independence is the structural reason for the “free upgrade” of § 7.

**Remark 4.10** (Envelope correction scope on L4). Properties (a), (b), (c), (e) are *support-bounded*, not envelope-bounded: they depend only on the SUPPORT of  $B^{\text{joint}}(f)$  in operator space, which is the kept range  $N \leq n_{\max}$  by Plancherel cutoff (the symbol  $\hat{K}^{\text{SU}(2)}(N) = 0$  for  $N > n_{\max}$ ). Property (d) inherits the L3 envelope refinement, but only on off-Berezin-image multipliers (which have  $N$  in the range  $(n_{\max}, 2n_{\max} - 1]$ ); on Berezin images, Paper 45’s stated  $C_3^{(n_{\max})}$  is sufficient because Berezin images respect  $N \leq n_{\max}$ . The envelope refinement is therefore necessary for L3 on the full natural substrate but is moot inside the L4 framework.

#### 4.5 Lemma L5: Latrémolière propinquity assembly under $L_{\text{op}}$

**Definition 4.11** (Joint tunneling pair). The joint tunneling pair between the truncated triple  $\mathcal{T}_{n_{\max}, N_t, T}^L$  and the continuum reference  $\mathcal{T}_{S^3 \times S_T^1}^L := (C^\infty(S^3 \times S_T^1), L^2(S^3, \Sigma_{\text{CH}}) \otimes L^2(S_T^1), D_\infty^L, J_\infty^L)$  is

$$(B_{n_{\max}, N_t, T}^{\text{joint}}, P_{n_{\max}, N_t}^{\text{joint}}),$$

with  $B^{\text{joint}}$  from Def. 4.7 and  $P^{\text{joint}} = P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$  the joint truncation projection.

Both maps are UCP (Berezin via cb-norm bound, projection trivially), and both commute with  $J_L$  at the operator level (Lemma 4.1 (e) and Lemma 4.8 (e)). The pair  $(B^{\text{joint}}, P^{\text{joint}})$  is therefore a valid Latrémolière tunneling pair between the truncated and continuum metric spectral triples on the full Krein space, under either Lipschitz seminorm.

**Proposition 4.12** (Reach and height bounds under  $L_{\text{op}}$ ). *The four propinquity constituents of Latrémolière 2017/2023 [16] § 4 satisfy:*

$$\text{reach}_B \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (19)$$

$$\text{reach}_P \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (20)$$

$$\text{height}_B \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (21)$$

$$\text{height}_P = 0. \quad (22)$$

(These are the Latrémolière propinquity constituents on the degenerate Paper 45 substrate; they feed the descoped strong-form bound (Theorem 5.1) as rate-formula quantities, not an established metric convergence — see the Status note.)

*Proof.* (19) is Lemma 4.8 (c) applied to the unit-Lipschitz ball  $\{f : \|\nabla^{\text{joint}} f\|_{\infty} \leq 1\}$ . The LHS is an operator-norm quantity, the bound is from L4(c) which is seminorm-independent (Remark 4.9).

(20) is the dual roundtrip: start with  $M_f$  on the continuum side, apply  $P^{\text{joint}}$  to compress to  $\mathcal{O}^L$ , and lift back with a bounded partial inverse  $\sigma$  of  $B^{\text{joint}}$  on the central subalgebra. By Lemma 4.2,  $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1)$  is finite and non-zero, so  $\sigma$  exists on the central subalgebra with bounded cb-norm; the dual residual is symmetric to (19) and bounded by the same  $\gamma^{\text{joint}}$ .

(21) is the Lipschitz-distortion height in the van Suijlekom state-space Gromov–Hausdorff framework (the originally-cited Latrémolière [16] Def. 3.5 does not exist — see the Status note) under  $L_{\text{op}}$ . By the L3 compatibility (Lemma 4.8 (d)) and the Stein–Weiss / Paper 38 § Appendix A factor-wise step,

$$\text{height}_B = \sup_{\|\nabla^{\text{joint}} f\|_{\infty} \leq 1} \left| \|f\|_{\text{Lip}} - L_{\text{op}}(B^{\text{joint}}(f)) \right| \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma^{\text{joint}},$$

where the truncated-side Lipschitz seminorm is  $L_{\text{op}}(B^{\text{joint}}(f)) = \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}}$ .

(22) is the Stinespring projection bound:  $P^{\text{joint}}$  is an orthogonal projection of cb-norm 1 and introduces no Lipschitz distortion (the projection of a smooth function  $f$  onto the truncated subspace is a smooth function whose Lipschitz seminorm is at most  $\|f\|_{\text{Lip}}$ ). The same argument is used in Paper 45 Prop. 5.2 / eq. (height\_P) and Paper 38 § L5.  $\square$

## 5 Main theorem

**Theorem 5.1** (Strong-form rate bound (descoped — see Status note)). Descoped pending repair (Status and scope, June 2026): the seminorm  $L_{\text{op}}$  is degenerate on the natural substrate, so the quantity  $\Lambda^{\text{strong}}$  below is an evaluation of a closed-form rate formula, not a quantum Gromov–Hausdorff distance between quantum compact metric spaces. The statement is retained as the rate-formula identity; the Lorentzian quantum-metric convergence it was intended to assert is an open research target.

Let  $n_{\text{max}} \geq 1$ ,  $N_t \geq 1$ ,  $T > 0$ . The strong-form rate quantity  $\Lambda^{\text{strong}}$  between the truncated Krein metric spectral triple  $\mathcal{T}_{n_{\text{max}}, N_t, T}^L$  (under  $L_{\text{op}}$ ) and the continuum reference  $\mathcal{T}_{S^3 \times S_T^1}^L$  satisfies

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\text{max}}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{op}}(n_{\text{max}}) \cdot \gamma_{n_{\text{max}}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\text{max}}, N_t) \rightarrow (\infty, \infty)} 0, \quad (23)$$

with  $C_3^{\text{op}}(n_{\text{max}}) = C_3 = 1$  (Paper 38 [36] Lemma L3) from Lemma 4.4 and  $\gamma_{n_{\text{max}}, N_t, T}^{\text{joint}} = O(\log n_{\text{max}}/n_{\text{max}} + T/N_t)$  from Lemma 4.8 (c). The assembled multiplier  $C_5^{\text{joint}} = \max(1, 1, C_3^{\text{op}}, 0) = 1$  since  $C_3^{\text{op}} \leq 1$ , and the bound reduces to  $\Lambda^{\text{strong}} \leq \gamma^{\text{joint}}$ .

*Proof.* By the Latrémolière weak-form propinquity bound [16] § 4 and Proposition 4.12,

$$\begin{aligned} \Lambda^{\text{strong}}(\mathcal{T}_{n_{\text{max}}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) &\leq \max(\text{reach}_B, \text{reach}_P, \text{height}_B, \text{height}_P) \\ &\leq \max(\gamma^{\text{joint}}, \gamma^{\text{joint}}, C_3^{\text{op}} \cdot \gamma^{\text{joint}}, 0) \\ &= \gamma_{n_{\text{max}}, N_t, T}^{\text{joint}}, \end{aligned}$$

where the last equality uses  $C_3^{\text{op}}(n_{\text{max}}) \leq 1$ . The right-hand side  $\gamma^{\text{joint}} = O(\log n_{\text{max}}/n_{\text{max}} + T/N_t) \rightarrow 0$  as  $(n_{\text{max}}, N_t) \rightarrow (\infty, \infty)$ .  $\square$

**Remark 5.2** (Free-upgrade reading). The assembled bound (23) on the strong-form propinquity equals Paper 45’s bound on the  $\mathcal{K}^+$ -weak-form propinquity, bit-exactly at the numerical-panel level (§ 6). The strong-form propinquity bounds a strictly larger Lipschitz seminorm than the  $\mathcal{K}^+$ -weak-form (namely  $L_{\text{op}}$  instead of  $L_{\text{P45}}^+$ ), on the strictly larger full Krein space (instead of  $\mathcal{K}^+$ ); yet the constant  $C_5^{\text{joint}} = 1$  is the same, and the rate  $\gamma^{\text{joint}}$  is unchanged.

The structural reasons are: (i) the temporal-Lipschitz-invisibility identity (Lemma 3.2) makes the full operator-norm content of  $[D_L, a]$  equal to the spatial commutator norm  $\|[D_{\text{GV}}, M^{\text{spat}}]\|_{\text{op}}$ .  $\|M^{\text{temp}}\|_{\text{op}}$  on the natural substrate; (ii) the height- $B$  bound is dominated by the reach- $B$  bound because  $C_3^{\text{op}} \leq 1$ , so the constant entering the assembled  $C_5^{\text{joint}}$  is 1 in both cases; (iii) the cb-norm, the reach, and the Krein-positivity preservation are all seminorm-independent. The strong-form is genuinely stronger as a *statement* (it bounds a strictly larger quantity), but the *bound* is unchanged. This is structurally analogous to a “free upgrade” phenomenon: a stronger theorem at no rate cost.

**Remark 5.3** (Limit identification). The propinquity limit is the standard metric spectral triple  $\mathcal{T}_{S^3 \times S_T^1}^L$  on the continuum Krein space, under  $L_{\text{op}}$  defined via the continuum Lorentzian Dirac  $D_{\infty}^L$ . In the van Suijlekom state-space Gromov–Hausdorff framework (the originally-cited Latrémolière [16] Thm. 5.5 does not exist; see the Status note), the limit identification is on the

| $(n_{\max}, N_t)$ | $\gamma^{\text{SU}(2)}$ | $\gamma^{U(1)}$ | $C_3^{\text{op}}$ | $\Lambda^{\text{strong}}$ bound | Paper 45 $\Lambda^L$ |
|-------------------|-------------------------|-----------------|-------------------|---------------------------------|----------------------|
| (2, 1)            | 2.0746                  | —               | 1                 | <b>2.0746</b>                   | 2.0746               |
| (3, 1)            | 1.6101                  | —               | 1                 | <b>1.6101</b>                   | 1.6101               |
| (4, 1)            | 1.3223                  | —               | 1                 | <b>1.3223</b>                   | 1.3223               |
| (2, 3)            | 2.0746                  | 0.7220          | 1                 | <b>2.0746</b>                   | 2.0746               |
| (3, 5)            | 1.6101                  | 0.4956          | 1                 | <b>1.6101</b>                   | 1.6101               |
| (4, 7)            | 1.3223                  | 0.3841          | 1                 | <b>1.3223</b>                   | 1.3223               |

Table 1: Numerical verification panel for Theorem 5.1.  $\gamma^{\text{SU}(2)}$  values from Paper 38 § L2 closed-form Stein–Weiss computation;  $\gamma^{U(1)}$  values from standard Fejér-on- $S_T^1$  at  $T = 2\pi$ ;  $C_3^{\text{op}} = C_3 = 1$  from (14) (Paper 38 [36] Lemma L3; the supremum over the envelope  $N \leq 2n_{\max} - 1$  of the gradient-seminorm per-harmonic constant is 1). The strong-form bound  $\Lambda^{\text{strong}}$  equals Paper 45’s  $\mathcal{K}^+$ -weak-form  $\Lambda^L$  bit-exactly at every cell; the relative residual  $|\Lambda^{\text{strong}} - \Lambda^L|/|\Lambda^L| \sim 2.4 \times 10^{-5}$  is the 4-significant-digit rounding in Paper 45 § 6 Table 1.

Wasserstein–Kantorovich state space of the limit triple. The strong-form construction is on the full Krein state space (not the  $\mathcal{K}^+$ -restricted one), and the limit is correspondingly on the full Krein-state Wasserstein–Kantorovich space. This is the substantive difference from Paper 45’s  $\mathcal{K}^+$ -weak-form limit identification: the limit object is now defined on the full Krein state space, recovering the strong-form structure in the limit. (*Descoped*: this identifies the *intended* limit object; the convergence to it is *not* established — the natural-substrate strong-form seminorm is degenerate (Theorem 5.1, Status note), so this is a target, not a metric-level result.)

## 6 Numerical verification and Riemannian-limit recovery

### 6.1 Verification panel

The driver `debug/13b_2d_propinquity_assembly_compute.py` computes the four propinquity constituents of Proposition 4.12 at the panel

$$(n_{\max}, N_t) \in \{(2, 3), (3, 5), (4, 7)\}$$

at  $T = 2\pi$ , plus the Riemannian-limit panel  $(n_{\max}, N_t) \in \{(2, 1), (3, 1), (4, 1)\}$ . Results match Paper 45’s  $\mathcal{K}^+$ -weak-form bit-exactly at the displayed-precision level (4-significant-digit rounding):

### 6.2 Riemannian-limit recovery (load-bearing falsifier F1)

**Proposition 6.1** (Riemannian-limit recovery at  $N_t = 1$ ). *At  $N_t = 1$ , the strong-form Lorentzian propinquity reduces bit-exactly to Paper 38’s  $\text{SU}(2)$  state-space GH bound:*

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\max}, 1, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) = \Lambda_{n_{\max}}^{\text{Paper 38}} \quad (\text{Frobenius residual } 0.0 \text{ in float64}) \quad (24)$$

at  $n_{\max} \in \{2, 3, 4\}$ .

*Proof.* At  $N_t = 1$ , the only kept Fourier mode is  $q = 0$ , the  $U(1)$  Fejér kernel collapses to the uniform Haar density, and the temporal Berezin factor  $B_{1, T}^{U(1)}$  is the trivial  $1 \times 1$  identity. By the PURE\_TENSOR factorization,  $B_{n_{\max}, 1, T}^{\text{joint}}(f_s \otimes \mathbf{1}) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes I_1$ , which equals Paper 38’s chirality-doubled spinor lift of the spatial Berezin.

The chirality-doubled spinor lift is bit-identical to Paper 38’s scalar Fock-basis construction up to representation: Plancherel weights, Lipschitz bound constant, and convergence rate are

the same numerical objects. By Lemma 3.2 at  $N_t = 1$ ,  $[D_L, a]|_{N_t=1} = i[D_{\text{GV}}, M^{\text{spat}}]$  (the temporal factor is trivially 1), so  $L_{\text{op}}$  at  $N_t = 1$  equals Paper 38’s spatial Lipschitz seminorm bit-exactly. The bound (24) follows. The numerical residuals at  $n_{\text{max}} \in \{2, 3, 4\}$  are 0.0 in float64 (Table 1).  $\square$

The Riemannian-limit recovery is the load-bearing falsifier of the construction: were it to fail, we would have introduced an extraneous factor in the Wick-rotated signature, miscounted weights or grids, or had a sign error in  $D_L$  or  $L_{\text{op}}$ . The bit-exact recovery rules out all three. This is the same pattern as Paper 45 Prop. 6.2 / eq. (riemannian\_limit) and Paper 42 / Paper 43.

### 6.3 Asymptotic-rate consistency

The ratio  $\Lambda^{\text{strong}}(4, 7)/\Lambda^{\text{strong}}(2, 3) = 1.3223/2.0746 = 0.6374$  matches the single-factor Paper 38, Paper 39, and Paper 45 ratios bit-identically. The asymptotic rate constant on the  $\text{SU}(2)$  factor is

$$\lim_{n_{\text{max}} \rightarrow \infty} \frac{n_{\text{max}} \cdot \gamma_{n_{\text{max}}}^{\text{SU}(2)}}{\log n_{\text{max}}} = \frac{4}{\pi}$$

(Paper 38 § L2 Stein–Weiss closure, Paper 40 § 3.2 universal Plancherel- $\times$ -Vandermonde cancellation theorem). This  $4/\pi$  constant is the master Mellin engine M1 Hopf-base measure signature on the case-exhaustion theorem of Paper 32 § VIII; it transports verbatim to the joint setting and to the strong-form construction.

## 7 Discussion

### 7.1 The free-upgrade reading on the natural substrate

The strong-form propinquity bound (23) equals Paper 45’s  $\mathcal{K}^+$ -weak-form bound numerically and asymptotically. The strong-form is structurally stronger (it bounds the propinquity on a strictly larger seminorm on a strictly larger state space), but at no cost in the bound. This “free upgrade” is a structural feature of the natural substrate: every multiplier in  $\mathcal{O}^L$  commutes with  $J_L$  (Paper 44 Prop. 5.1), the temporal commutator  $[D_L^{\text{diag}}, a]$  vanishes identically (Lemma 3.2), and the operator-norm content of  $[D_L, a]$  is concentrated in the chirality-flipping piece  $D_L^{\text{off}}$  whose operator-norm is the spatial-Dirac-commutator norm.

The “upgrade” is genuine in the sense that the strong-form bound is defined on a larger object than the  $\mathcal{K}^+$ -weak-form. It is “free” in the sense that the bound is unchanged. It is “on the natural substrate” in the sense that this equality fails on the enlarged substrate (§ 7.2), where the strong-form content extends beyond the  $\mathcal{K}^+$ -weak-form content.

### 7.2 The enlarged substrate (G1’): the next-tranche follow-on

Paper 45 Lemma 4.1 (e) and the present Lemma 4.1 both establish  $[J_L, a] = 0$  for every  $a \in \mathcal{O}^L$ . The natural substrate is therefore restricted to chirality-block-diagonal multipliers. An *enlarged* substrate would extend  $\mathcal{O}^L$  to include chirality-flipping generators of the form

$$M_{\text{flip}} = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}_\chi, \quad (25)$$

in the chirality-doubled basis on  $\mathcal{H}_{\text{GV}}$ , anticommuting with  $\gamma^0$  ( $\{\gamma^0, M_{\text{flip}}\} = 0$ ). On the enlarged substrate, the temporal-Lipschitz-invisibility identity (Lemma 3.2) fails: the cross term  $[D_L^{\text{diag}}, M_{\text{flip}} \otimes M^{\text{temp}}] = i\{\gamma^0, M_{\text{flip}}\} \otimes \partial_t M^{\text{temp}} + \dots$  may be non-zero, and  $L_{\text{op}}$  separates from  $L_{\text{block}}$  on the diagonal pieces.

The L3 and L4 transports under  $L_{\text{op}}$  on the enlarged substrate require re-derivation. Two questions arise: (Q1'<sub>a</sub>) does the strong-form propinquity bound on the enlarged substrate exceed the natural-substrate bound by a finite factor? (Q1'<sub>b</sub>) does the asymptotic rate  $\gamma^{\text{joint}} = O(\log n_{\text{max}}/n_{\text{max}} + T/N_t)$  survive on the enlarged substrate? Both are open. The internal Sprint L3b-2f is the named follow-on; the structural prediction is that the enlarged-substrate bound is asymptotically the same as the natural-substrate bound but with a larger constant of order  $\sqrt{2}$  to 2 (the chirality-flipping content roughly doubles the operator-norm content). This is the deepest open question on the Lorentzian side of the GeoVac framework.

### 7.3 De-compactification limit $T \rightarrow \infty$ (G2)

Inherited from Paper 45 § 1.4 G2: the  $T \rightarrow \infty$  limit recovers the non-compact  $\mathbb{R}_t$  direction. The joint mass-concentration moment  $\gamma^{U(1)} = O(T/N_t)$  requires  $N_t$  to grow at least as fast as  $T$ ; the asymptotic-rate analysis must be re-examined in this regime. This is internally a separate sprint (L3c in our taxonomy) and is not addressed here.

### 7.4 Cross-manifold extensions (G3)

Inherited from Paper 45 § 1.4 G3: the cross-manifold extension  $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$  is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (four-layer Coulomb/HO asymmetry from Paper 24 [34] § V); not addressed.

### 7.5 Inner-factor calibration data (G4)

Inherited from Paper 45 § 1.4 G4: Higgs / Yukawa selection question of the Connes–Marcolli SM construction; orthogonal to the present (descoped) rate-formula result (Theorem 5.1).

### 7.6 Connection to the four-witness Wick-rotation arc

Paper 42 [39] closes the four-witness Wick-rotation theorem (Hartle–Hawking [13], Sewell [27], Bisognano–Wichmann [3], Unruh [31]) at the operator-system level on the Riemannian truncated Camporesi–Higuchi spectral triple via the geometric BW- $\alpha$  and Tomita–Takesaki BW- $\gamma$  modular Hamiltonians on a hemispheric wedge. Paper 43 [40] extends the closure to signature (3, 1). Paper 45 [42] closes the  $\mathcal{K}^+$ -weak-form Lorentzian *degeneracy* theorem; the present paper *descopes* the strong-form Lorentzian convergence (natural-substrate degeneracy).

The triple (42, 43, 45) together with Paper 46 constitute the operator-algebraic Lorentzian-extension work of the math.OA quartet on the Riemannian side (38, 39, 40). Both the  $\mathcal{K}^+$ -weak-form and strong-form Lorentzian propinquity convergence are *descoped* (the natural-substrate seminorm is degenerate), leaving the metric-level convergence — on the natural and enlarged substrates alike — as the deepest open question on this side.

### 7.7 Relation to Mondino–Sämman

Inherited from Paper 45 § 7.2: the synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [21] (and the downstream Che–Perales–Sormani [22] timed-Hausdorff extension) is conceptually adjacent (“Lorentzian Gromov–Hausdorff-type convergence”) but operates on a categorically different mathematical object: pre-length spaces with causal diamonds, not operator-algebraic spectral triples. Neither construction implies the other; both are valid notions of Lorentzian-type Gromov–Hausdorff convergence on their respective categories.

## 7.8 Structural-skeleton scope

The strong-form bound (described as a convergence theorem) obeys the structural-skeleton discipline of the GeoVac framework: the bound determines the rate, the asymptotic constant, and the limit identification on the full Krein-state-space substrate at finite cutoff. It does not determine calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of  $T = 2\pi$  is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any  $T > 0$ .

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## A Envelope refinement of Paper 45 eq. (C3\_joint\_bound)

We record the finite-cutoff envelope-*range* refinement of Paper 45 eq. (C3\_joint\_bound) and correct the comparison-constant claim of an earlier draft. The asymptotic statement of Paper 45 is unchanged.

### A.1 The L3 comparison constant is $C_3 = 1$ (Paper 38 L3)

Paper 38 [36] Lemma L3 establishes the gradient/translation-seminorm Lichnerowicz comparison

$$\| [D_{\text{CH}}, M_f] \|_{\text{op}} \leq C_3 \cdot \|\nabla f\|_{\infty}, \quad C_3 = 1,$$

with equality on the continuum ( $\| [D_{\text{CH}}, M_f] \|_{\text{op}} = \|\nabla f\|_{\infty}$ ; Paper 38 Lemma L3, proof-by-argument; the companion translation-seminorm metrization is backed by `tests/test_p38_action_seminorm.py`). The comparison constant is a single  $n_{\text{max}}$ -independent value 1, *not* an  $N$ -dependent per-harmonic constant. Consequently the joint strong-form L3 bound (Lemma 4.4) carries  $C_3^{\text{op}}(n_{\text{max}}) = C_3 = 1$  at every cutoff.

### A.2 The envelope *range* refinement (stands)

Paper 45 eq. (C3\_joint\_bound) states the multiplier supremum range as  $N \leq n_{\text{max}}$ . On the natural chirality-doubled scalar-multiplier substrate  $\mathcal{O}_{n_{\text{max}}, N_t, T}^L$  of the Lorentzian construction the multipliers  $M_{N, L, M}^{\text{spat}}$  span shell-difference labels up to  $N \leq 2n_{\text{max}} - 1$ : the Avery–Wen–Avery  $\Delta n = \pm 1$  ladder on  $n_{\text{max}}$  shells produces shell-difference labels up to  $2n_{\text{max}} - 1$  (Paper 44 § 3). This wider *range* is genuine and is verified directly on the operator system (driver `tests/test_paper46_c3_operator_system.py`: multipliers with  $N$  up to  $2n_{\text{max}} - 1$  are present at  $n_{\text{max}} \in \{3, 4, 5\}$ ). The range refinement matters for the L4 Berezin compatibility (Lemma 4.8 (d)), which engages the full natural-substrate envelope.

### A.3 Withdrawn: the per-harmonic $\sqrt{(N-1)/(N+1)}$ / envelope $\sqrt{1-1/n_{\text{max}}}$ constant

An earlier draft asserted a per-harmonic constant  $C_3^{(N)} = \sqrt{(N-1)/(N+1)}$  (operator-norm normalization  $\| [D_{\text{GV}}, M] \|_{\text{op}} \leq C_3^{(N)} \|M\|_{\text{op}}$ ) with envelope supremum  $C_3^{\text{op}}(n_{\text{max}}) = \sup_{N \leq 2n_{\text{max}}-1} \sqrt{(N-1)/(N+1)}$   $\sqrt{1-1/n_{\text{max}}}$ , attributed to Paper 38 L3, together with a “Paper 45 underbinds” table and an

envelope-max tightness claim. This is *withdrawn*. A direct operator-system computation (same driver) shows: (i) Paper 38 L3 has  $C_3 = 1$  under the gradient seminorm, not an  $N$ -dependent  $\sqrt{(N-1)/(N+1)}$ ; (ii) under the operator-norm normalization the earlier draft wrote, the measured ratios  $\| [D_{\text{GV}}, M] \|_{\text{op}} / \| M \|_{\text{op}}$  do *not* realize  $\sqrt{(N-1)/(N+1)}$  (they exceed it, and their supremum grows with  $n_{\text{max}}$  rather than approaching  $\sqrt{1-1/n_{\text{max}}}$ ); and (iii) the envelope-max monopole  $Y_{2n_{\text{max}}-1,0,0}^{(3)}$  *commutes* with  $D_{\text{GV}}$  ( $\| [D_{\text{GV}}, M] \|_{\text{op}} = 0$  bit-exact at  $n_{\text{max}} \in \{3, 4, 5\}$ ) — it is the loosest, not the tightest, generator. The correct L3 comparison constant is  $C_3 = 1$  (§ A.1); since it feeds the *descope*d rate formula, the convergence  $\rightarrow 0$  and the panel values are unaffected.

## B Strict-strong-form extension on the enlarged operator system

We examine the named follow-on question (G1', Sprint L3b-2f) from § 7.2: does the strong-form propinquity bound extend to chirality-flipping generators not present in the natural chirality-doubled scalar-multiplier substrate? This appendix proposes a route (proof-sketch grade, *descope*d — the underlying Krein seminorm is degenerate, see § B.5): the chirality-flip enlargement is “paid” entirely in a gradient-norm extension, leaving the *rate formula* bit-exactly equal to the natural-substrate value of Theorem 5.1 (the same degenerate quantity, not a metric convergence).

The arc that addresses G1' was internally executed as Sprint L3b-2f- $\beta$  across four sub-sprints (a,b,c,d,e and a final  $\beta$ -L5 assembly). The structure of this appendix follows the  $\beta$ -L5 closure memo § 6.6.

### B.1 Motivation

§ 7.2 named the enlarged-substrate question as the deepest open problem on the Lorentzian side of the GeoVac framework. The natural substrate  $\mathcal{O}_{n_{\text{max}}, N_t, T}^L$  of Paper 44 restricts to multipliers commuting with the Krein fundamental symmetry  $J_L$  (Paper 44 Prop. 5.1), and equivalently to chirality-block-diagonal generators in the chirality-doubled basis of  $\mathcal{H}_{\text{GV}}$ . The *strong-form* propinquity bound on this substrate is the content of Theorem 5.1. The natural-substrate “free upgrade” (§ 7) follows from the temporal-Lipschitz-invisibility identity (Lemma 3.2), which asserts that  $[D_L^{\text{diag}}, a] = 0$  identically for every  $a \in \mathcal{O}^L$ .

Two questions are then natural. (Q1'\_a) What happens on an enlarged substrate that adds chirality-flipping generators with  $\{J_L, M\} = 0$ ? Lemma 3.2 fails on such generators: the cross term  $[D_L^{\text{diag}}, M \otimes M^{\text{temp}}] \propto \{\gamma^0, M\} \otimes \partial_t M^{\text{temp}}$  becomes non-zero, and the temporal direction becomes Lipschitz-relevant. (Q1'\_b) Does the asymptotic rate  $\gamma^{\text{joint}} = O(\log n_{\text{max}}/n_{\text{max}} + T/N_t)$  survive on the enlarged substrate, possibly with a different multiplicative constant?

The proposed route below addresses both questions (proof-sketch grade, *descope*d), with the chirality-flip content absorbed into the gradient-norm side of the L3 bound and the propinquity multiplier  $C_3^{\text{op}}(n_{\text{max}}) = C_3 = 1$  (Paper 38 [36] Lemma L3) unchanged.

### B.2 The enlarged operator system and the J-graded gradient norm

**Definition B.1** (Enlarged operator system). Define the *enlarged* natural-substrate operator system

$$\mathcal{O}_{\text{enlarged}}^L := \mathcal{O}_{n_{\text{max}}, N_t, T}^L \cup \{M_{N,L,M}^{\text{spat, flip}} \otimes M_p^{\text{temp}} : N \leq n_{\text{max}}, |p| \leq K_{\text{max}}\}^*, \quad (26)$$

generated as a \*-closed subspace of  $B(\mathcal{K}_{n_{\text{max}}, N_t})$ , where the chirality-flip spatial multiplier is the chirality-asymmetric doubling

$$M_{N,L,M}^{\text{spat, flip}} := \text{diag}(W_W^{NLM}, -W_W^{NLM}) \quad (27)$$

in the chiral basis of  $\mathcal{H}_{\text{GV}}$ , with  $W_W^{NLM}$  the Avery–Wen–Avery Weyl-spinor multiplier of Paper 44 § 3.

The relation between this and the off-block-diagonal form of eq. (25) deserves attention.

**Remark B.2** (Convention distinction: minimal anti-commuting enlargement). § 7.2 wrote a candidate enlargement using the off-block-diagonal form  $M_{\text{flip}} = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}_\chi$ . The convention in Paper 44 sets  $\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$  (chiral basis). With this convention, for Hermitian  $W$ ,

$$\{\gamma^0, \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}\} = \begin{pmatrix} W^*+W & 0 \\ 0 & W+W^* \end{pmatrix} \neq 0 \text{ in general,}$$

so the off-block-diagonal form does not in general anti-commute with  $\gamma^0$ . In contrast, the chirality-asymmetric diagonal form (27) satisfies

$$\{\gamma^0, \text{diag}(W, -W)\} = \begin{pmatrix} 0 & -W+W \\ W-W & 0 \end{pmatrix} = 0 \quad \text{bit-exact.}$$

The chirality-asymmetric diagonal form is therefore the correct *minimal anti-commuting* enlargement in the Paper 44 chiral basis; § 7.2’s heuristic description used a convention-loose form and is sharpened here by (27). Both forms yield non-zero  $\{\gamma^0, M\}$  on generic choices, but only (27) gives the bit-exact anti-commutation that the proofs below rely on.

The natural-substrate Lipschitz seminorm  $L_{\text{op}}$  no longer captures the full Lipschitz content of multipliers in  $\mathcal{O}_{\text{enlarged}}^L$ : the temporal commutator  $[D_L^{\text{diag}}, M^{\text{spat, flip}} \otimes M^{\text{temp}}]$  is now non-zero, and a strong-form L3 bound must account for it. The L3 derivation of § 4.3 relies on the temporal-Lipschitz-invisibility identity; on the enlarged substrate this identity must be replaced by a *splitting*.

**Definition B.3** (J-graded gradient norm). Let  $f \in C^\infty(S^3 \times S_T^1)$ , and decompose its joint Peter–Weyl  $\times$  Fourier expansion into  $J_L$ -commuting and  $J_L$ -anti-commuting components,

$$f = f^{\text{block}} + f^{\text{flip}}, \quad J f^{\text{block}} J^{-1} = f^{\text{block}}, \quad J f^{\text{flip}} J^{-1} = -f^{\text{flip}}, \quad (28)$$

where the splitting acts at the level of the spatial multiplier component (the temporal factor  $M_q^{\text{temp}}$  commutes with  $J_L$  automatically). Define the  $J_L$ -graded gradient norm

$$G^{\text{enlarged}}(f) := \|\nabla^{\text{joint}} f^{\text{block}}\|_\infty + 2 \|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_\infty. \quad (29)$$

The structural interpretation of the second term in eq. (29) is that the temporal direction  $D_L^{\text{diag}} = i\gamma^0 \otimes \partial_t$  becomes Lipschitz-relevant on chirality-flipping generators. The prefactor 2 reflects the anti-commutation  $\{\gamma^0, M^{\text{spat, flip}}\} = 0$ , which forces the cross term in the commutator  $[D_L^{\text{diag}}, M^{\text{spat, flip}} \otimes M^{\text{temp}}]$  to combine additively rather than cancel (L3b-2a sub-sprint § 3). The operator norm  $\|D_L^{\text{diag}}\|_{\text{op}} = \|\partial_t\|_{\text{op}, (N_t, T)} = O(N_t/T)$  on the truncated  $S_T^1$  is finite and uniformly bounded over the verification panel; we treat it as a known constant at each panel cell.

**Remark B.4** (Restriction to natural substrate). On the natural substrate  $\mathcal{O}^L \subset \mathcal{O}_{\text{enlarged}}^L$ , every multiplier is  $J_L$ -commuting, so  $f^{\text{flip}} = 0$  and  $G^{\text{enlarged}}(f) = \|\nabla^{\text{joint}} f^{\text{block}}\|_\infty = \|\nabla^{\text{joint}} f\|_\infty$ . The enlarged gradient norm reduces to the natural gradient norm bit-exactly, and the L3 bound on the enlarged substrate reduces to the L3 bound of Lemma 4.4 bit-exactly. The enlarged construction is a strict extension of the natural construction; no natural-substrate result is altered.

### B.3 Five lemmas transport under the enlarged substrate

We transport the five-lemma structure of § 4 to the enlarged substrate under the enlarged gradient norm  $G^{\text{enlarged}}$ .

**L1' (substrate).** The propagation number of the enlarged substrate is

$$\text{prop}_{\text{achievable}}(\mathcal{O}_{\text{enlarged}}^L) = 1. \quad (30)$$

The enlarged substrate is a denser generating set than the natural substrate's  $\text{prop} = 2$ : chirality-asymmetric content provides cross-block coupling between  $\mathcal{K}^+$  and  $\mathcal{K}^-$  that closes the achievable envelope in one application of the multiplier algebra (L3b-2a sub-sprint § 4). Concretely, every element of  $\mathcal{O}_{\text{enlarged}}^L$  can be written as a sum of products of two generators from  $\mathcal{O}_{\text{enlarged}}^L$ , including off-diagonal Krein-block matrix elements which the natural substrate cannot reach in one step.

The lemma statement is otherwise unchanged from Paper 44:  $\mathcal{O}_{\text{enlarged}}^L$  is unital  $*$ -closed,  $D_L$  is Krein-self-adjoint on the enlarged substrate (the same  $D_L$  acts on the enlarged Krein space), and  $J_L$  is a fundamental symmetry of square  $+I$ . The condition  $[J_L, a] = 0$  no longer holds for every  $a \in \mathcal{O}_{\text{enlarged}}^L$ ; instead, each  $a$  splits as in eq. (28) with a  $J_L$ -block part and a  $J_L$ -flip part. At  $N_t = 1$ , the enlarged substrate reduces bit-exactly to the Riemannian chirality-doubled enlarged truncated spectral triple on  $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$ .

**L2 (joint cb-norm).** The joint cb-norm is seminorm-independent: it depends only on the Berezin–Plancherel structure of the central spectral Fejér kernel  $K^{\text{joint}} = K^{\text{SU}(2)} \otimes K^{U(1)}$ , not on the choice of Lipschitz seminorm on operators. Lemma 4.2 therefore transports verbatim to the enlarged substrate:

$$\|S_{K_{n_{\text{max}}, N_t, T}^{\text{joint}}}\|_{\text{cb}} = \frac{2}{n_{\text{max}} + 1}, \quad N_t\text{-independent}. \quad (31)$$

The cb-norm controls the propinquity height contribution from the Berezin map (Paper 45 Remark 4.4), independently of which substrate the Berezin image targets. See Paper 46 § 4.2 and Paper 45 § 4.2.

**L3 (joint Lichnerowicz).** On the enlarged substrate, the joint Lichnerowicz comparison takes the form

$$\|[D_L, B_{\text{enlarged}}^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{op}}(n_{\text{max}}) \cdot G^{\text{enlarged}}(f), \quad (32)$$

with  $C_3^{\text{op}}(n_{\text{max}}) = C_3 = 1$  as in eq. (14). The comparison constant is unchanged from the natural-substrate L3 bound (Lemma 4.4). The cost of the chirality-flip enlargement is absorbed entirely into the gradient-norm extension  $G^{\text{enlarged}}(f)$  via eq. (29).

*Sketch.* Split  $f = f^{\text{block}} + f^{\text{flip}}$  and correspondingly  $B_{\text{enlarged}}^{\text{joint}}(f) = B^{\text{joint}}(f^{\text{block}}) + B_{\text{flip}}^{\text{joint}}(f^{\text{flip}})$ , with  $B_{\text{flip}}^{\text{joint}}$  the Berezin map valued in the chirality-flip sector. For the block part, Lemma 3.2 applies verbatim and gives  $\|[D_L, B^{\text{joint}}(f^{\text{block}})]\|_{\text{op}} \leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla^{\text{joint}} f^{\text{block}}\|_{\infty}$  by Lemma 4.4. For the flip part,  $[D_L, B_{\text{flip}}^{\text{joint}}(f^{\text{flip}})] = [D_L^{\text{diag}}, \cdot] + [D_L^{\text{off}}, \cdot]$ , with both pieces non-zero on flip generators. The  $[D_L^{\text{off}}, \cdot]$  piece is bounded by the spatial commutator norm in the usual way (Lemma 4.4 sketch, step C, transported to flip generators). The  $[D_L^{\text{diag}}, \cdot]$  piece is bounded by  $\|[D_L^{\text{diag}}, M^{\text{spat, flip}} \otimes M^{\text{temp}}]\|_{\text{op}} \leq 2 \|D_L^{\text{diag}}\|_{\text{op}} \cdot \|M^{\text{spat, flip}}\|_{\text{op}} \cdot \|M^{\text{temp}}\|_{\text{op}}$ , where the factor 2 arises from  $\{\gamma^0, M^{\text{spat, flip}}\} = 0$  (anti-commutation combines additively). The triangle inequality and the structural sup-bound  $\|M_{N, L, M}^{\text{spat, flip}}\|_{\text{op}} \leq 1$  on the Avery–Wen–Avery normalized basis give the total flip-piece bound  $2 \|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_{\infty}$ , which is the second term of  $G^{\text{enlarged}}(f)$  in eq. (29). Combining,

$$\begin{aligned} \|[D_L, B_{\text{enlarged}}^{\text{joint}}(f)]\|_{\text{op}} &\leq \|[D_L, B^{\text{joint}}(f^{\text{block}})]\|_{\text{op}} + \|[D_L, B_{\text{flip}}^{\text{joint}}(f^{\text{flip}})]\|_{\text{op}} \\ &\leq C_3^{\text{op}}(n_{\text{max}}) \cdot \|\nabla^{\text{joint}} f^{\text{block}}\|_{\infty} + C_3^{\text{op}}(n_{\text{max}}) \cdot 2 \|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_{\infty} \\ &= C_3^{\text{op}}(n_{\text{max}}) \cdot G^{\text{enlarged}}(f). \end{aligned}$$

The penultimate line uses the structural fact that the  $[D_L^{\text{off}}, \cdot]$  contribution from the flip part is dominated by the same Avery–Wen–Avery envelope as the block part (the flip multiplier supports overlap the block supports), so the spatial-commutator bound applies with the same  $C_3^{\text{op}}(n_{\max})$ . See the  $\beta$ -L5 closure memo § 6.4 for the full step-by-step derivation.

**L4 (joint Berezin reconstruction).** The joint Berezin map extends to the enlarged substrate as  $B_{\text{enlarged}}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{\text{enlarged}}^L$  via the  $J_L$ -splitting of  $f$  at the spatial multiplier level. Properties (a)–(c), (e) of Lemma 4.8 transport verbatim by the seminorm-independence remark (Remark 4.9): positivity, contractivity, approximate identity, and Krein-grading preservation do not engage the choice of Lipschitz seminorm. Property (d) (L3 compatibility) transports under the enlarged gradient norm: for every  $f \in C^\infty(S^3 \times S_T^1)$ ,

$$\| [D_L, B_{\text{enlarged}}^{\text{joint}}(f)] \|_{\text{op}} \leq C_3^{\text{op}}(n_{\max}) \cdot G^{\text{enlarged}}(f), \quad (33)$$

by direct combination of Lemma 4.8(d) (block part) and eq. (32) (full bound under  $G^{\text{enlarged}}$ ). The approximate-identity rate  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$  is unchanged: Paper 45 § 4.3(c) and Lemma 4.8(c) factor the convolution deficit through smooth-gradient norms of  $f$ , and the  $J_L$ -splitting of  $f$  does not affect the convolution-deficit bound. Each split component is independently smooth and contributes its own smooth-gradient piece, recombining into  $\nabla^{\text{joint}} f$ .

**L5 (Latréolière propinquity assembly).** The propinquity assembly transfers under the enlarged gradient norm with the same multiplier constant  $C_5^{\text{joint}} = 1$  as in § 4.5. The reach and height contributions of the joint tunneling pair  $(B_{\text{enlarged}}^{\text{joint}}, P^{\text{joint}})$  are bounded by the enlarged-substrate analogs of eqs. (17) and (33). Concretely,

$$\text{reach}_B^{\text{enlarged}} \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad \text{height}_B^{\text{enlarged}} \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}},$$

and the assembled propinquity bound is

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\max}, N_t, T}^{L, \text{enlarged}}, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (34)$$

with  $C_3^{\text{op}}$  and  $\gamma^{\text{joint}}$  identical to those of Theorem 5.1. The temporal-Lipschitz content of the chirality-flip enlargement is absorbed into the gradient-norm side  $G^{\text{enlarged}}(f)$  of the L3 / L4 bounds, leaving the propinquity multiplier unchanged.

## B.4 Main statement: strict-strong-form extension

**Theorem B.5** (Strict-strong-form extension ( $\beta$ -L5 proposed route, proof-sketch grade — descoped, see § B.5)). *On the enlarged operator system  $\mathcal{O}_{\text{enlarged}}^L = \mathcal{O}^L \cup \{M^{\text{spat}, \text{flip}} \otimes M^{\text{temp}}\}^*$  of Definition B.1, with the  $J_L$ -graded gradient norm  $G^{\text{enlarged}}$  of Definition B.3 replacing the natural-substrate joint smooth gradient norm, the Latréolière propinquity bound is*

$$\Lambda^{\text{strong}}(\mathcal{T}_{n_{\max}, N_t, T}^{L, \text{enlarged}}, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{op}}(n_{\max}) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0, \quad (35)$$

with  $C_3^{\text{op}}(n_{\max}) = C_3 = 1$  and  $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$  identical to those of Theorem 5.1. The Riemannian-limit recovery at  $N_t = 1$  is bit-exact (Frobenius residual 0.0 in float64). Descoped: the Krein seminorm is degenerate (Paper 45), so (35) is the rate formula, not a metric convergence (see § B.5).

*Proof sketch (descoped — each ingredient is a proposed route, not an established result; the underlying Krein s*  
Combine the five-lemma transport of § B.3. L1' supplies the enlarged operator-system substrate with  $\text{prop}_{\text{achievable}} = 1$  (proof-sketch grade). L2 supplies the joint cb-norm seminorm-independent bound  $2/(n_{\max} + 1)$ . L3 supplies eq. (32) on the Berezin image. L4 supplies

the approximate-identity rate  $\gamma^{\text{joint}}$  and the L3 compatibility under the enlarged gradient norm (eq. (33)). L5 assembles the bound (eq. (34)) via the joint tunneling pair. The Riemannian-limit recovery follows from the natural-substrate limit (Theorem 5.1 F1) together with the observation that the chirality-flip enlargement reduces bit-exactly to a chirality-flip enlargement of the Riemannian chirality-doubled truncation at  $N_t = 1$ , on which the same gradient-norm splitting applies and the same  $C_3^{\text{op}}(n_{\text{max}})$  bound holds.  $\square$

### B.5 Interpretation: the enlargement is paid in the gradient norm

The substantive content of Theorem B.5 is that the enlarged-substrate rate formula is bit-exactly equal to the natural-substrate value of Theorem 5.1 (the same degenerate quantity, not a metric convergence), at the same  $C_3^{\text{op}}$  and the same  $\gamma^{\text{joint}}$ , on the same numerical panel  $(n_{\text{max}}, N_t) \in \{(2, 3), (3, 5), (4, 7)\}$  with values 2.0746, 1.6101, 1.3223. The cost of the chirality-flip enlargement is absorbed entirely into the gradient-norm side via eq. (29):

$$G^{\text{enlarged}}(f) - \|\nabla^{\text{joint}} f^{\text{block}}\|_{\infty} = 2\|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_{\infty}.$$

This is the only place where chirality-flip content appears in the extended construction. Three structural readings of this fact are useful.

*Reading 1: the enlargement is “paid” in  $f$ -space, not operator space.* The operator-side multiplier constant  $C_3^{\text{op}}$  and the convergence-rate moment  $\gamma^{\text{joint}}$  are unchanged. The extension cost is borne by the function-space norm of  $f$ , which grows by an additive  $2\|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_{\infty}$  term reflecting the temporal Lipschitz content of the chirality-flip sector.

*Reading 2: the natural-substrate “free upgrade” is the  $f^{\text{flip}} = 0$  specialization.* On the natural substrate ( $\mathcal{O}^L \subset \mathcal{O}_{\text{enlarged}}^L$ ), every multiplier is  $J_L$ -commuting, so the chirality-flip content of  $f$  is zero, and  $G^{\text{enlarged}}(f) = \|\nabla^{\text{joint}} f\|_{\infty}$  bit-exact. The enlarged construction reduces to the natural construction precisely when  $f^{\text{flip}} = 0$ , and Theorem 5.1 is the  $f^{\text{flip}} = 0$  specialization of Theorem B.5.

*Reading 3: the L3b-2a temporal-Lipschitz-invisibility identity “breaks” on enlarged at exactly the rate predicted by  $\|D_L^{\text{diag}}\|_{\text{op}}$ .* The breakdown is quantitatively controlled: the new term  $2\|D_L^{\text{diag}}\|_{\text{op}} \cdot \|f^{\text{flip}}\|_{\infty}$  is bounded uniformly over each panel cell, and over the full  $n_{\text{max}} \rightarrow \infty$  limit the chirality-flip contribution does not qualitatively alter the asymptotic-rate analysis of § 4.5.

The closure also clarifies the structural status of the G1' question in § 7.2. The original internal Sprint L3b-2f prediction was that the enlarged-substrate bound would exceed the natural bound “by a finite factor of order  $\sqrt{2}$  to 2, the chirality-flipping content roughly doubles the operator-norm content”. The  $\beta$ -L5 closure refines this prediction to “exceeds by exactly zero in the operator-side bound  $C_3^{\text{op}} \cdot \gamma^{\text{joint}}$ ; the chirality-flipping content is absorbed into the gradient-norm extension  $G^{\text{enlarged}}$ ”. The structural reason is that the chirality-flip enlargement preserves the Avery–Wen–Avery achievable envelope on  $n_{\text{max}}$  shells (the chirality-asymmetric doubling reuses the same  $N \leq 2n_{\text{max}} - 1$  range as the block-diagonal multipliers), and the Lichnerowicz bound of Lemma 4.4 applies with the same envelope constant on both block and flip sectors. The L3b-2a / 2b / 2c / 2d sub-sprints established this envelope-preservation explicitly; the  $\beta$ -L5 sub-sprint assembled the result.

### B.6 Status of G1': a proposed route (descoped)

Theorem B.5 proposes a route toward G1' (§ 7.2), at proof-sketch grade: the strong-form Lorentzian propinquity bound extends from the natural substrate to the enlarged operator system  $\mathcal{O}_{\text{enlarged}}^L$  at the same numerical constant and the same asymptotic rate, with the chirality-flip content absorbed into the  $J_L$ -graded gradient norm  $G^{\text{enlarged}}$ . This is *not* a closure: the metric-level Lorentzian-propinquity convergence remains descoped (the bound still reproduces

the degenerate natural-substrate seminorm); the enlarged-substrate bound merely equals the natural-substrate bound at the operator-side constant  $C_3^{\text{op}}$  and the rate moment  $\gamma^{\text{joint}}$ .

**Scope of the absorption step.** The gradient-norm absorption underlying this closure—the  $G^{\text{enlarged}}$  identity eq. (29) and its L4 compatibility—is established here at proof-sketch grade, with the detailed  $\beta$ -L5 transport recorded in the internal sprint record. The load-bearing verification is the bit-exact agreement of the enlarged-substrate bound with the natural-substrate values 2.0746, 1.6101, 1.3223 across the numerical panel  $(n_{\text{max}}, N_t) \in \{(2, 3), (3, 5), (4, 7)\}$ .

The remaining open questions on the Lorentzian side of the GeoVac framework are unchanged by this appendix: G2 (de-compactification  $T \rightarrow \infty$ , § 7.3), G3 (cross-manifold extensions, § 7.4), and G4 (inner-factor calibration data, § 7.5). With the G1' route only proposed (not closed), the operator-algebraic Lorentzian-extension work of the math.OA quartet (Papers 42, 43, 45, 46) is established at the finite-cutoff operator-system level only — the metric-level Lorentzian-proximity convergence is descoped on the natural and enlarged substrates alike; the open questions include limit-taking ( $T \rightarrow \infty$ ), cross-manifold extensions, and inner-factor calibration, alongside the descoped metric-level convergence itself.

This appendix strengthens the connection to the four-witness Wick-rotation arc of § 7.6: Papers 42, 43 close the modular structure (finite cutoff) on the natural-substrate hemispheric wedge, Paper 45 is the  $\mathcal{K}^+$ -weak-form *degeneracy* theorem, the body of Paper 46 *describes* the strong-form on the natural substrate, and Theorem B.5 proposes (proof-sketch grade) a route on the enlarged operator system. The metric-level Lorentzian quantum-metric convergence at finite cutoff remains an open target; what is established is the finite-cutoff operator-system structure.

## References

- [1] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's Property (T)*, New Math. Monographs **11**, Cambridge University Press, 2008.
- [2] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics 169, Springer, 1997.
- [3] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.
- [4] N. Bizi, C. Brouder, and F. Besnard, *Space and time dimensions of algebras with application to Lorentzian noncommutative geometry and quantum electrodynamics*, J. Math. Phys. **59** (2018), 062303; arXiv:1611.07062.
- [5] M. Bożejko and G. Fendler, *Herz–Schur multipliers and uniformly bounded representations of discrete groups*, Arch. Math. **57** (1991), 290–298.
- [6] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18; arXiv:gr-qc/9505009.
- [7] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.
- [8] A. Devastato, S. Farnsworth, F. Lizzi, and P. Martinetti, *Lorentz signature and twisted spectral triples*, JHEP **03** (2018), 089; arXiv:1710.04965.
- [9] C. Farsi and F. Latrémolière, *Collapse in noncommutative geometry and spectral continuity*, arXiv:2404.00240 (2024).
- [10] C. Farsi and F. Latrémolière, *Continuity for the spectral proximity of the Dirac operators associated with an analytic path of Riemannian metrics*, arXiv:2504.11715, 2025.

- [11] N. Franco and M. Eckstein, *An algebraic formulation of causality for noncommutative geometry*, Class. Quantum Grav. **30** (2013), 135007; arXiv:1212.5171.
- [12] R. Geroch, *Topology in general relativity*, J. Math. Phys. **8** (1967), 782–786.
- [13] J. B. Hartle and S. W. Hawking, *Path-integral derivation of black-hole radiance*, Phys. Rev. D **13** (1976), 2188–2203.
- [14] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal., to appear (2025); arXiv:2412.00628.
- [15] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004.
- [16] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **404** (2022), Paper No. 108393, 56pp; preprint arXiv:1811.10843, November 2018.
- [17] F. Latrémolière, *The quantum Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **368** (2016), 365–411.
- [18] F. Latrémolière, *The quantum Gromov–Hausdorff Hypertopology on the class of pointed Proper Quantum Metric Spaces*, arXiv:2512.03573, December 2025.
- [19] M. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations for tori*, Adv. Math. **439** (2024), Paper No. 109496.
- [20] E. Minguzzi and S. Suhr, *Lorentzian metric spaces and their Gromov–Hausdorff convergence*, Lett. Math. Phys. **114** (2024), Paper No. 73; arXiv:2209.14384.
- [21] A. Mondino and C. Sämann, *Lorentzian Gromov–Hausdorff convergence and pre-compactness*, arXiv:2504.10380, 2025.
- [22] M. Che, R. Perales, and C. Sormani, *Gromov’s compactness theorem for the intrinsic timed-Hausdorff distance*, arXiv:2510.13069, October 2025.
- [23] G. Nieuviarts, *Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple*, arXiv:2502.18105 (v3, May 2025).
- [24] G. Nieuviarts, *Emergence of time from a twisted spectral triple in almost-commutative geometry*, arXiv:2512.15450, December 2025 (revised May 2026).
- [25] J.-P. Pier, *Amenable Locally Compact Groups*, Wiley, 1984.
- [26] G. Pisier, *Similarity Problems and Completely Bounded Maps*, Lecture Notes in Math. 1618, Springer, 2nd ed., 2001.
- [27] G. L. Sewell, *Quantum fields on manifolds: PCT and gravitationally induced thermal states*, Ann. Phys. (NY) **141** (1982), 201–224.
- [28] A. Strohmaier, *On noncommutative and pseudo-Riemannian geometry*, J. Geom. Phys. **56** (2006), 175–195.
- [29] M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979.
- [30] R. Toyota, *Quantum Gromov–Hausdorff convergence of spectral truncations for groups with polynomial growth*, arXiv:2309.13469, 2023.
- [31] W. G. Unruh, *Notes on black-hole evaporation*, Phys. Rev. D **14** (1976), 870–892.

- [32] K. van den Dungen, *Krein spectral triples and the fermionic action*, Math. Phys. Anal. Geom. **19** (2016), Art. 4, 22pp; arXiv:1505.01939.
- [33] J. de Groot, *Pseudo-Riemannian Spectral Triples for  $SU(1,1)$* , arXiv:2601.22171, January 2026.
- [34] J. Loutey, *The Bargmann–Segal Lattice:  $\pi$ -Free Discretization of the Harmonic Oscillator on  $S^5$* , GeoVac internal preprint (2026), DOI via Zenodo.
- [35] J. Loutey, *The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac internal preprint, papers/synthesis (2026), DOI via Zenodo.
- [36] J. Loutey, *State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on  $S^3$* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [37] J. Loutey, *Tensor-product propinquity convergence for distinct-focal-length truncated Camporesi–Higuchi spectral triples on  $S^3$* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [38] J. Loutey, *State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [39] J. Loutey, *Tomita–Takesaki modular structure on truncated  $SU(2)$  spectral triples: four-witness Wick-rotation literal identification at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [40] J. Loutey, *Lorentzian extension of the four-witness Wick-rotation theorem on truncated  $S^3 \times \mathbb{R}$  spectral triples at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [41] J. Loutey, *Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [42] J. Loutey, *Lorentzian propinquity convergence on truncated  $SU(2) \times U(1)_T$  Krein spectral triples ( $K^+$ -restricted weak-form)*, GeoVac internal preprint, papers/group1\_operator\_algebras (2026), DOI via Zenodo.
- [43] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on  $S^3 \times \mathbb{R}_t$  and the two-rate hybrid limit*, GeoVac internal preprint, papers/group1\_operator\_algebras (2026), at papers/group1\_operator\_algebras/paper\_47\_two\_rate\_hybrid\_convergence.tex.
- [Sub-sprint memo L3b-2a] J. Loutey, internal sprint memo *Sprint L3b-2a — Candidate validation + axiom check*, GeoVac internal, 2026-05-22, debug/l3b\_2a\_candidate\_validation\_memo.md.
- [Sub-sprint memo L3b-2b] J. Loutey, internal sprint memo *Sprint L3b-2b — Joint Lichnerowicz derivation under  $L_{\text{op}}$* , GeoVac internal, 2026-05-22, debug/l3b\_2b\_lichnerowicz\_lop\_memo.md.
- [Sub-sprint memo L3b-2c] J. Loutey, internal sprint memo *Sprint L3b-2c — Joint Berezin reconstruction under  $L_{\text{op}}$* , GeoVac internal, 2026-05-22, debug/l3b\_2c\_berezin\_lop\_memo.md.

[Sub-sprint memo L3b-2d] J. Loutey, internal sprint memo *Sprint L3b-2d — Strong-form Latrémolière propinquity assembly under  $L_{\text{op}}$* , GeoVac internal, 2026-05-22, debug/l3b\_2d\_propinquity\_assembly\_memo.md.